

Predicting cryptocurrency returns with machine learning: Evidence from high-dimensional factor modeling

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Abstract: We investigate the predictability of cryptocurrency returns using a comprehensive set of macroeconomic and cryptocurrency-specific factors and a set of 12 machine learning models. To enhance interpretability, we employ SHAP analysis to quantify the marginal contribution of each factor to model outputs. We further assess the economic value of predictive signals by constructing long-short and long-only portfolios. Empirically, tree-based methods, particularly random forests, deliver the highest predictive accuracy and outperform neural network and linear benchmarks, with predictability substantially stronger than that documented in equity markets. Across models, the market-to-realized-value ratio, new addresses, and active addresses consistently emerge as the most influential predictors, with higher values associated with higher expected returns. Portfolio results show that neural network-based strategies achieve the highest cumulative performance, indicating meaningful investment gains. Overall, our findings demonstrate the value of machine learning for return forecasting in the cryptocurrency market and provide practical insights for investors and financial analysts operating in highly volatile and evolving cryptocurrency environments.

Keywords: Cryptocurrency return predicting; Machine learning; SHAP interpretability; Cross-sectional factors; Portfolio performance

JEL classification: C52; C55; C58; G12; G17

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1. Introduction

Cryptocurrencies have attracted substantial attention from investors, regulators, and the media ever since the introduction of Bitcoin by Nakamoto (2008). Functioning as peer-to-peer electronic cash systems, cryptocurrencies facilitate direct online payments between parties without the need for intermediaries such as financial institutions. This distinctive characteristic sets them apart from the majority of traditional financial assets, as they lack ties to higher authorities and physical representation and are infinitely divisible. Unlike conventional financial assets, the value of cryptocurrencies is not anchored in tangible assets, a country's economy, or a specific firm. Instead, their valuation is rooted in the security of an algorithm that traces all transactions. The widespread adoption of cryptocurrencies can be attributed to low transaction costs, a peer-to-peer system, and government-free design. Consequently, this surge in adoption has led to increased trading volume, heightened volatility, and notable fluctuations in the prices of cryptocurrencies.

In the field of cryptocurrencies, studies have focused mainly on cryptocurrency market efficiency. One branch of literature has investigated cryptocurrency's price efficiency. Urquhart (2016) is noteworthy for being the first to demonstrate the inefficiency of the Bitcoin market, a conclusion subsequently supported by Zhang et al. (2018). However, contrasting findings emerge from the studies of Tiwari et al. (2018), who argue that Bitcoin returns exhibit market efficiency. Grobys and Sapkota (2019) state that the cryptocurrency market is efficient. Schilling and Uhlig (2019) propose that the trajectory of cryptocurrency prices should adhere to a martingale, suggesting a lack of predictability in cryptocurrency returns.

Another branch of literature has examined return predictability in the cryptocurrency market. In this branch, some studies endeavor to identify risk factors that can predict or explain cryptocurrency returns. Bianchi and Babiak (2021) and Borri et al. (2022) examine risk premiums in a cross-section of cryptocurrencies. The former identifies liquidity, size, reversal, market, and downside risks as the main driving factors, while the latter asserts that macroeconomic risk is a priced element in the cryptocurrency market. Within dynamic valuation frameworks, both Cong et al. (2021) and Sockin and Xiong (2023) find that cryptocurrency returns can be forecasted by incorporating momentum and investor attention. Biais et al. (2023) argue that the fundamental value of a cryptocurrency lies in its stream of net transactional benefits, depending on its future prices. Liu et al. (2022) draw the conclusion that market, size, and momentum factors effectively capture cross-sectional expected cryptocurrency returns. Adding to this discourse, Liu and Tsyvinski (2021) demonstrate that network factors influence cryptocurrency returns, with momentum as a critical determinant of future cryptocurrency returns. Additionally, Yae and Tian (2022) forecast the returns of Bitcoin, Ethereum, and Ripple using correlation with stock markets. Bouri et al. (2022) employ the regression filter techniques to predict the returns of the top five cryptocurrencies. Babiak et al. (2022) find that the interaction between lagged returns and trading volume contains substantial predictive information for understanding the dynamics of cryptocurrency returns. Cai and Zhao (2024) examine the salience effect in the cryptocurrency market and identify it as a significant risk factor in explaining cryptocurrency returns.

In recent years, some studies have departed from conventional models and instead embraced machine learning to predict the prices or returns of Bitcoin and other cryptocurrencies. McNally et al. (2018) employ a long short-term memory (LSTM) model to predict Bitcoin prices, while Jakubik et al. (2023) use LSTM networks to extract unstructured information from financial news and subsequently use machine learning models to enhance Bitcoin price forecasting and trading. Liu et

al. (2021) adopt a stacked denoising autoencoders approach to predict Bitcoin prices. Chowdhury et al. (2020) employ neural networks, K-nearest neighbors, and gradient boosting trees to forecast the Cryptocurrency Index 30 and 9 cryptocurrencies included in this index. Sun et al. (2020) leverage the light gradient boosting machine method for predicting the prices of 41 cryptocurrencies within a given market. Akyildirim et al. (2021) explore the prediction of price trends for the 12 most liquid cryptocurrencies using logistic regression, support vector machines, artificial neural networks, and random forests. Liu et al. (2023) utilize both ordinary least squares regression and the extreme gradient boosting model to predict cryptocurrency returns. Bouteska et al. (2024) forecast 4 popular cryptocurrencies through the application of ensemble learning and deep learning models. It is noteworthy that these studies mainly focus on major popular cryptocurrencies and rely on a few machine learning methods. This underscores a gap in the existing literature, suggesting a need for more comprehensive investigations that encompass a diverse range of cryptocurrencies and employ a broader range of machine learning methodologies.

In addition to cryptocurrency prediction, a body of literature has been dedicated to the portfolio diversification of cryptocurrencies. Some studies delve into the role of Bitcoin in well-diversified portfolios. Wu and Pandey (2014) argue that Bitcoin provides the potential to enhance the performance of an investor's portfolio. Briere et al. (2015) further posit that the inclusion of Bitcoin significantly improves the risk-adjusted returns of portfolios. Kajtazi and Moro (2019) provide additional support, suggesting that Bitcoin enhances portfolio performance by increasing returns rather than reducing risk. Notably, Platanakis and Urquhart (2020) demonstrate substantial benefits by adding Bitcoin to a traditional portfolio, revealing significantly higher risk-adjusted returns. Other studies explore the diversification benefits of cryptocurrencies. Platanakis et al. (2018) find little difference between naïve and optimal diversification for four popular cryptocurrencies. Liu (2019) emphasizes the potential advantages of diversifying across ten major cryptocurrencies, significantly improving investment results. Platanakis and Urquhart (2019) show that in the cryptocurrency market, the Black-Litterman model exhibits superior out-of-sample risk-adjusted returns and lower risks compared to naïve and Markowitz models. Additionally, Brauneis and Mestel (2019) find that combining cryptocurrencies enriches the spectrum of low-risk cryptocurrency investment opportunities. However, the studies mentioned above only examine Bitcoin or a small number of cryptocurrencies over a short period, relying on in-sample estimation and utilizing traditional portfolio techniques. This implies the necessity for more expansive examinations that span a diverse array of cryptocurrencies and leverage advanced machine learning techniques.

This study makes significant contributions to the existing literature on cryptocurrency return prediction. First, we pioneer the construction of a comprehensive factor set comprising 8 macroeconomic and 17 characteristic factors. This unique dataset captures the multifaceted nature of the cryptocurrency landscape by spanning diverse economic and crypto-intrinsic dimensions. Second, we employ a comprehensive battery of 12 machine learning models alongside 2 linear models, representing the most extensive approach in the literature. Through the analysis of the out-of-sample predictability and factor importance, we identify tree models as superior in predictive performance and pinpoint specific factors exerting substantial influence on returns. Third, we leverage the SHAP (SHapley Additive exPlanations) method, a novel approach for black-box model interpretability, to delve into the impact of cryptocurrency factors on model output. This enables us to explain the decision-making processes within machine learning models and identify the factors

that positively or negatively contribute most to the predictive outcomes. Fourth, we construct both long-short and long-only cryptocurrency portfolios informed by the models and factors. Evaluating the out-of-sample performance of these portfolios and examining the dynamics of their cumulative returns provide a holistic understanding of the practical implications and real-world efficacy of our framework.

The remainder of this paper is organized as follows. In Section 2, we describe our data and the methodologies employed for prediction. We present our empirical analysis in Section 3. We look at the out-of-sample predictability and discuss which factors matter most. We also examine the impact of factors on model output. Section 4 analyzes portfolios based on model predictabilities and discusses portfolio gains. In Section 5, we conclude our study. Detailed discussions of the methods used and factor construction are in the Appendix.

2. Data and methodology

2.1 Data

We collect daily trading data of all cryptocurrencies available from Coinmarketcap.com. This website aggregates information from over 200 major exchanges and provides comprehensive data, including daily opening and closing prices, high and low prices, market capitalization, and trading volumes for most cryptocurrencies. For each cryptocurrency on the website, its price is computed via a volume-weighted average of individual exchange prices. To alleviate concerns about survivorship bias, Coinmarketcap.com lists both active and defunct cryptocurrencies.

We utilize daily closing prices of cryptocurrencies to calculate corresponding weekly returns.¹ We require that cryptocurrencies have complete information on price, market capitalization, and trading volume. In addition, we exclude small-cap cryptocurrencies with market capitalizations below \$10,000,000. The selected sample period for trading data spans from January 1, 2014, to March 31, 2023. The choice of starting on January 1, 2014, is driven by the limited liquidity and trading volume of cryptocurrencies before this date.

We gather fundamental data on characteristic factors of cryptocurrencies from App.intheblock.com, a platform that provides exclusive paid data for a wide range of cryptocurrencies. The platform collaborates with over 200 leading companies in the industry, including Coinmarketcap.com. These data are used to construct characteristic factors. The trading data of financial markets is obtained from the WIND database, while Google search data is collected from the Google Trends website. The search data is initially presented in daily format and is subsequently transformed into weekly data to align with the trading dates of cryptocurrencies.

Our cryptocurrency factors are composed of macroeconomic and characteristic factors.² Based on the fundamental data, we build 17 cryptocurrency characteristic factors spanning 8 dimensions, including network, financial, ownership, exchange, and social. The details of these predictors are provided in the Appendix. These various dimensions can encapsulate distinct attributes of cryptocurrencies, each of which can exert an influence on the returns of cryptocurrencies. We also construct 8 macroeconomic factors by uncertainty indices, market indices, commodity prices, and

¹ We predict cryptocurrencies' weekly returns since the fundamental data available is only weekly.

² Inspired by Gu et al. (2020), we classify cryptocurrency factors into characteristic factors and macroeconomic factors.

cryptocurrency index. The construction of the cryptocurrency index is provided in the Appendix. Table 1 summarizes the macroeconomic and characteristic factors.

Table 1

Cryptocurrency factors.

Characteristic factors	
Dimension 1: Network	Change rates of new addresses (NA), active addresses (AA), zero balance addresses (ZBA), active addresses ratio (AAR), number of transactions (NT), and transactions volume (TV)
Dimension 2: Finance	Change rates of market value to realized value ratio (MRR) and network value to transaction ratio (NTR)
Dimension 3: Ownership	Change rates of large holders netflow (LHN), large holders inflow (LHI), and large holders outflow (LHO)
Dimension 4: Exchange	Change rates of net flows (NF) and total flow (TF)
Dimension 5: Social	Change rates of Bitcoin's Google Trends (GT)
Dimension 6: Market	Market excess return (RMRF)
Dimension 7: Size	Small minus big (SMB)
Dimension 8: Momentum	Up minus down (UMD)
Macroeconomic factors	
Uncertainty indices	Returns of EPU and Twitter-based economic uncertainty (TEU) indices
Financial market indices	Returns of S&P500 (SPX), volatility index (VIX), and US dollar index (USDIX)
Commodity prices	Returns of the gold price (GLD) from COMEX and the crude oil price (WTI) from NYMEX
Cryptocurrency index	Return of Google Trends cryptocurrency index (GTCX) derived by using Google search volume of five keywords (Bitcoin, BTC, cryptocurrency, blockchain, and crypto)

Table 2 presents the descriptive statistics for cryptocurrency factors. Large holders inflow (LHI) and large holders outflow (LHO) are notable for their large means, with only net flows (NF) exhibiting a negative mean. Most characteristic factors demonstrate considerable standard deviations, especially for large holders netflow (LHN) and NF. The number of transactions (NT), active addresses (AA), and active addresses ratio (AAR) stand out for their markedly high kurtosis. Notably, all factors show significant ADF values at the 1% level, indicating that they are stationary time series. Moreover, the correlations among cryptocurrency factors are provided in Figure 1, revealing significantly low correlation coefficients. Notably, the correlation between AA and AAR is exceptionally high at 0.91. AAR and NT have a moderate correlation of 0.74, while SPX and VIX exhibit a negative correlation of -0.39 .

Table 2

The descriptive statistics for cryptocurrency factors.

	Mean	Std	Min	Max	Skewness	Kurtosis	ADF
EPU	0.181	0.990	−0.848	14.425	8.597	114.114	−25.306
TEU	0.154	0.745	−0.941	24.098	8.293	208.271	−19.979
SPX	0.002	0.048	−1.000	1.967	−6.970	412.010	−14.119
VIX	0.020	0.187	−0.427	1.348	2.017	9.870	−32.810
USDXX	0.001	0.018	−0.391	0.679	11.474	423.751	−19.923
GLD	0.002	0.020	−0.087	0.102	−0.029	3.349	−28.253
WTI	0.008	0.067	−0.317	0.328	0.051	6.866	−26.747
GTCX	0.018	0.205	−0.615	1.365	1.325	4.427	−26.342
NA	0.239	2.414	−4.275	94.930	20.666	623.985	−33.848
AA	0.121	1.234	−1.000	57.080	21.946	749.032	−23.725
ZBA	0.218	1.794	−1.000	87.247	20.277	656.629	−23.789
AAR	0.101	1.143	−1.000	52.667	20.750	685.360	−25.344
NT	0.191	1.565	−1.000	99.461	20.913	830.092	−23.996
TV	0.779	4.278	−1.000	98.909	11.004	164.307	−18.765
MRR	0.017	0.616	−1.000	61.996	28.157	620.598	−16.271
NTR	0.947	5.238	−1.000	99.728	10.414	136.690	−17.787
LHI	1.064	5.368	−1.000	99.709	9.212	112.069	−23.722
LHO	1.028	5.402	−2.353	98.675	9.415	114.249	−19.806
LHN	0.148	10.593	−9.945	99.609	0.710	31.625	−37.739
NF	−0.400	8.010	−9.948	99.835	−0.336	57.514	−33.798
TF	0.623	4.265	−7.168	99.270	11.040	174.474	−17.813
GT	0.089	2.879	−1.000	100.000	20.609	521.744	−9.524
RMRF	0.029	0.955	−10.61	55.838	−6.065	541.834	−17.851
SMB	0.048	0.949	−9.308	60.047	−7.450	363.252	−20.835
UMD	0.110	0.667	−0.969	67.000	5.972	382.549	−14.260

Notes: The critical value at the 1% significance level is −3.431.

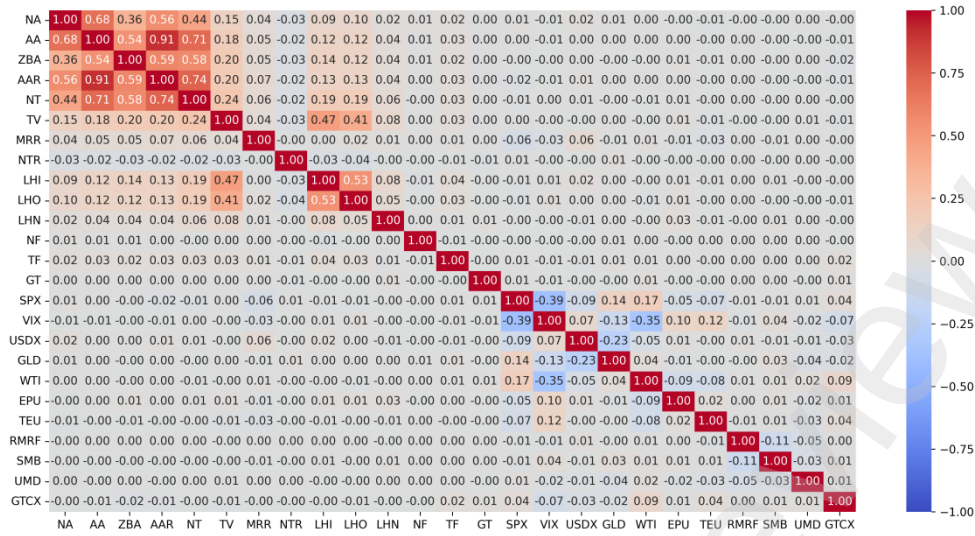


Figure 1. The correlation heatmap of cryptocurrency factors.

2.2 Methodology

Throughout our analysis, we utilize a general additive prediction error model to describe the relation between a cryptocurrency's return and its corresponding predictors,

$$r_{i,t+1} = \mathbb{E}_t[r_{i,t+1}] + \epsilon_{i,t+1}. \quad (1)$$

Furthermore, we assume the conditional expectation of cryptocurrency i 's return, $r_{i,t+1}$, given the information available at period t , is a constant function of a specified set of predictors:

$$\mathbb{E}_t[r_{i,t+1}] = g(z_{i,t}), \quad (2)$$

where $z_{i,t}$ is a P -dimensional vector of predictors, cryptocurrencies are indexed by $i = 1, 2, \dots, N$, weeks are indexed by $t = 1, \dots, T$. The functional form of $g(\cdot)$ is left unspecified. We aim to search for the prediction model that gives the best prediction performance from a set of candidates.

Our analysis incorporates 12 machine learning methodologies alongside 2 linear models. Specifically, we employ the ordinary least squares (OLS) regression, OLS using only market, size, and momentum (OLS-3), partial least squares (PLS), least absolute shrinkage and selection operator (LASSO), ridge (Ridge), elastic net (ENet), random forest (RF), gradient boosted regression trees (GBRT), K-nearest neighbor (KNN), and neural networks with one to five layers (NN1-NN5).

We follow the standard method in the literature for hyperparameter selection, model estimation, and performance evaluation. Specifically, we divide our data into three disjoint periods while maintaining chronological order: the training sample (2014-2017), the validation sample (2018-2020), and the testing sample (2021-2023). In the training sample, we estimate the model parameters subject to the prespecified hyperparameters for a given machine learning model. Subsequently, we utilize the validation sample to tune the hyperparameters of the machine learning models. The objective is to select the hyperparameters that minimize the loss function based on the observations in this sample. The testing sample serves the purpose of evaluating the predictive performance of the machine learning model. Notably, these data have not been used in either the model estimation or tuning, ensuring an unbiased assessment of the models' predictabilities. We standardize the inputs across the training, validation, and testing samples to facilitate comparability and robustness using the Z-score.

3. Empirical analysis

We commence our analysis by examining the prediction performance of all models through the out-of-sample R^2 and delve into predictability across different subsamples.

3.1 Out-of-sample predictability

To assess predictive performance for individual cryptocurrency return forecasts, we calculate the out-of-sample R^2 as

$$R_{oos}^2 = 1 - \frac{\sum_{(i,t) \in \mathcal{T}_3} (r_{i,t} - \hat{r}_{i,t})^2}{\sum_{(i,t) \in \mathcal{T}_3} r_{i,t}^2}, \quad (3)$$

where $\mathcal{T}_3$ indicates that fits are only assessed on the testing sample, whose data never enter model estimation or tuning; $\hat{r}_{i,t}$ is predicted weekly return. The results for different models and subsamples are summarized in Table 3.

3.1.1 Full sample analysis

When we include all cryptocurrencies, a majority of machine learning models exhibit excellent predictive capabilities. Notably, tree models, particularly RF, outperform other machine learning models. OLS achieves an R_{oos}^2 of 3.901%, suggesting that even a simple linear model retains certain predictability. It is noteworthy that the R_{oos}^2 of OLS-3 is significantly lower than that of OLS (0.044% vs. 3.901%), indicating that relying only on market, size, and momentum factors is insufficient to account for the entire predictive power in the linear model.

The regularized models do not demonstrate an improvement in their R_{oos}^2 compared to the linear models, suggesting that the factors used are all important and do not need dimension reduction. In the regularized models, both PLS and Ridge share an identical R_{oos}^2 (3.901%), mirroring that of OLS. However, both LASSO and ENet show no predictive power, recording zero R_{oos}^2 values.³

The tree models, RF and GBRT, exhibit formidable predictive capabilities that elevate the R_{oos}^2 to surpass 20%. Notably, RF slightly outperforms GBRT, boasting a higher R_{oos}^2 (25.265 vs. 24.911). Such improvement demonstrates the superiority of machine learning in capturing intricate interactions among predictors in the cryptocurrency market. This advantage aligns with similar findings for stock markets, as highlighted in both Gu et al. (2020) and Leippold et al. (2022).

Although the neural network models also raise their R_{oos}^2 to exceed 15%, they are inferior to the tree models. Among the five neural network models, NN4 demonstrates the highest R_{oos}^2 at 18.704%, suggesting that the model with four hidden layers has the best predictive performance. Note that increasing the number of hidden layers in neural networks tends to increase the R_{oos}^2 . However, this improvement is limited to models with three and five hidden layers, indicating that the benefits of deep learning do not extend to NN3 and NN5. Furthermore, the predictabilities of neural network models exhibit notable stability, with their R_{oos}^2 consistently ranging from 15% to 19%. KNN performs less effectively than tree and neural network models but exceeds the regularized and linear models. The R_{oos}^2 of KNN is 15.831%, slightly lower than NN1 (15.837 %).

In addition to the out-of-sample R^2 , we also employ the test of Diebold and Mariano (1995) to conduct comparisons of methods, evaluating the differences in out-of-sample predictive accuracy between two models. The results of the Diebold-Mariano test are presented in Table 4. Our findings

³ Indeed, the R_{oos}^2 values for both LASSO and ENet are nearly zero, i.e. -0.0000032733 .

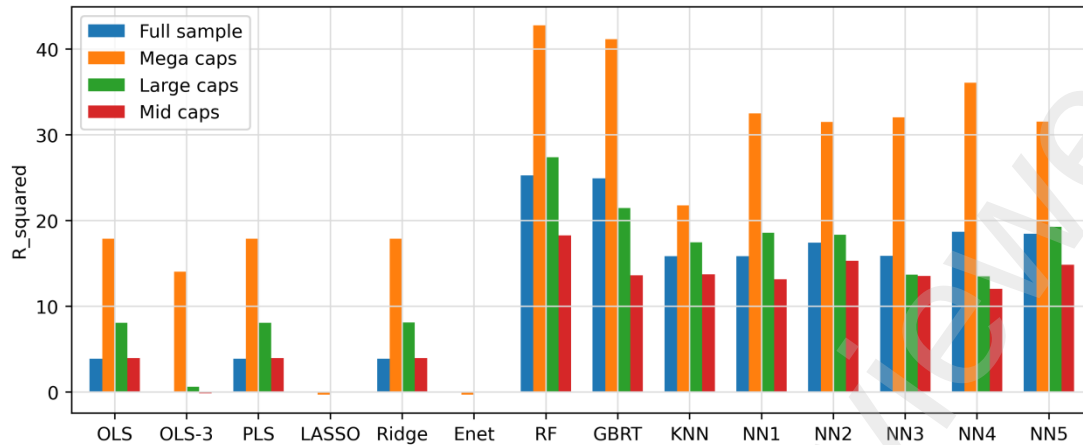
are consistent with those derived from the out-of-sample R^2 , indicating that utilizing the out-of-sample R^2 for model selection is robust in the cryptocurrency market.

It is noteworthy that the cryptocurrency market is driven by certain popular cryptocurrencies, typically categorized as mega-cap cryptocurrencies. The model predictability for mega-cap cryptocurrencies may differ from that for the full sample, as well as from large-cap and mid-cap cryptocurrencies. This motivates us to consider subgroup analysis.

Table 3

Weekly out-of-sample predictive R^2 in percentage.

	All	Mega-cap	Large-cap	Mid-cap
OLS	3.901	17.884	8.089	3.977
OLS-3	0.044	14.035	0.625	−0.145
PLS	3.901	17.884	8.089	3.977
LASSO	0.000	−0.308	0.000	−0.017
Ridge	3.901	17.887	8.100	3.977
ENet	0.000	−0.308	0.000	−0.017
RF	25.265	42.762	27.372	18.277
GBRT	24.911	41.146	21.476	13.624
KNN	15.831	21.762	17.467	13.726
NN1	15.837	32.488	18.593	13.170
NN2	17.435	31.495	18.352	15.326
NN3	15.877	32.032	13.706	13.544
NN4	18.704	36.090	13.503	12.053
NN5	18.478	31.527	19.265	14.855



Notes: This table reports weekly out-of-sample predictive R^2 of all models. The models considered include the ordinary least squares (OLS), OLS using only market, size, and momentum (OLS-3), partial least squares regression (PLS), least absolute shrinkage and selection operator (LASSO), ridge (Ridge), elastic network (ENet), random forest (RF), gradient boosted regression tree (GBRT), K-nearest neighbor (KNN), and neural networks with 1 to 5 layers (NN1-NN5). All the numbers are expressed as a percentage.

3.1.2 Mega-cap cryptocurrencies

To explore potential heterogeneity in the model predictability, we conduct subgroup analysis on mega-cap (cryptocurrencies with market capitalizations above \$1,000,000,000 each week), large-cap (cryptocurrencies with market capitalizations between \$100,000,000 and \$1,000,000,000 each week), and mid-cap (cryptocurrencies with market capitalizations between \$10,000,000 and \$100,000,000 each week).

Table 4

Comparison of weekly out-of-sample prediction using Diebold-Mariano test.

	OLS-3	PLS	LASSO	Ridge	ENet	RF	GBRT	KNN	NN1	NN2	NN3	NN4	NN5
OLS	_-0.677	0.000	_-0.684	0.000	_-0.684	3.748*	3.686*	2.093*	2.094*	2.374*	2.101*	2.597*	2.557*
OLS-3		0.677	_-0.008	0.677	_-0.008	4.425*	4.363*	2.770*	2.771*	3.051*	2.778*	3.274*	3.234*
PLS			_-0.684	0.000	_-0.684	3.748*	3.686*	2.093*	2.094*	2.374*	2.101*	2.597*	2.557*
LASSO				0.684	0.000	4.432*	4.370*	2.777*	2.778*	3.059*	2.785*	3.281*	3.242*
Ridge					_-0.684	3.748*	3.686*	2.093*	2.094*	2.374*	2.101*	2.597*	3.234*
ENet						4.432*	4.370*	2.777*	2.778*	3.059*	2.778*	3.281*	3.242*
RF							_-0.062	_-1.655	_-1.654	_-1.374	_-1.647	_-1.151	_-1.191
GBRT								_-1.593	_-1.592	_-1.312	_-1.585	_-1.089	_-1.129
KNN									0.001	0.281	0.008	0.504	0.464
NN1										0.280	0.007	0.503	0.463
NN2											_-0.273	0.223	0.183
NN3												0.496	0.456
NN4													_-0.040

Notes: This table reports pairwise Diebold-Mariano test statistics comparing the out-of-sample prediction performance based on the full sample among all models. Positive (negative) numbers indicate the column (row) model outperforms the row (column) model. Asterisks indicate the difference is significant at 5% level.

The third column of Table 3 summarizes the R^2_{oos} for mega-cap cryptocurrencies. The results suggest that all models have a much better predictive performance compared to the full sample. Linear models nearly double their performance, with OLS achieving a larger R^2_{oos} than OLS-3 (17.884% vs. 14.035%). This means that the inclusion of only three factors (market, size, and momentum) remains insufficient to explain all predictive power. Regularized models, including PLS and Ridge, experience an increase in R^2_{oos} (17.884% vs. 17.887%) to surpass 15%. It is worth noting that the R^2_{oos} of Ridge is slightly larger than that of OLS, indicating redundancy in certain factors for predicting weekly returns of mega-cap cryptocurrencies.

Tree and neural network models maintain their most dominance among machine learning models, yielding R^2_{oos} to above 40% and 30%, respectively. Notably, RF remains an advantage over GBRT and NN1-NN5, boasting the highest R^2_{oos} of 42.762%. GBRT follows closely with an R^2_{oos} of 41.146%. In neural network models, NN4 continues to exhibit exceptional predictive performance, recording the largest R^2_{oos} (36.090%). The improvement in R^2_{oos} is limited to the first four hidden layers. In addition, the predictive performance of neural network models is still stable, with R^2_{oos} ranging from 31% to 37%.

The R^2_{oos} of KNN stands at 21.762%, higher than linear and regularized models but significantly lower than tree and neural network models. Furthermore, it is observed that KNN's R^2_{oos} is much higher than that in the full sample. Finally, the Diebold-Mariano tests for mega-cap cryptocurrencies are presented in the Appendix. The corresponding results align with those obtained from the out-of-sample R^2 .

3.1.3 Large-cap cryptocurrencies

The R^2_{oos} for large-cap cryptocurrencies is presented in the fourth column of Table 3, indicating that most models have a better predictive performance compared to the full sample. However, their predictabilities are markedly inferior to those for mega-cap cryptocurrencies. The R^2_{oos} of OLS is 8.089%, which is twice that of the full sample but less than half of mega-cap cryptocurrencies. Surprisingly, the R^2_{oos} of OLS-3 experiences a substantial drop to 0.625%, still larger than the full sample. In regularized models, the R^2_{oos} of Ridge is slightly larger than that of PLS (8.100% vs. 8.089%), with both R^2_{oos} being twice that of the full sample but less than half of mega-cap cryptocurrencies.

Tree and neural network models are still excellent machine learning models, but their performance has dropped significantly compared to mega-cap cryptocurrencies. In tree models, the R^2_{oos} of RF consistently exceeds that of GBRT (27.372% vs. 21.476%), indicating the superiority of RF. The expanding advantage of RF over GBRT suggests an increasing disparity in their predictabilities. In neural network models, NN5 attains the largest R^2_{oos} of 19.265%, while NN4 records the smallest. This means that increasing hidden layers does not increase the R^2_{oos} . The R^2_{oos} of neural network models ranges from 13 to 20%, suggesting that the difference in their predictabilities increase slightly.

It is noteworthy that the predictability of KNN surpasses that of NN3 and NN4 for the first time. The R^2_{oos} of KNN stands at 17.467%, slightly smaller than NN1 and NN2. Finally, the Diebold-Mariano tests for large-cap cryptocurrencies are provided in the Appendix. The corresponding results align with those obtained from the out-of-sample R^2 .

3.1.4 Mid-cap cryptocurrencies

The last column of Table 3 shows the R^2_{oos} for mid-cap cryptocurrencies. The predictive performance of all models continues to decline. The R^2_{oos} of OLS (3.977%) is marginally larger than that of the full sample but only half of large-cap cryptocurrencies and one-fifth of mega-cap cryptocurrencies. Notably, the R^2_{oos} (−0.145%) of OLS-3 turns to negative for the first time, indicating that OLS using only three factors has no predictive power for mid-cap cryptocurrencies. In regularized models, PLS and Ridge maintain the same R^2_{oos} (3.977%), slightly larger than that of the full sample. The R^2_{oos} of LASSO and ENet (−0.017%) is slightly lower than that of the full sample.

Surprisingly, the predictive performance of tree and neural network models demonstrates a significant drop, even lower than that of the full sample. The R^2_{oos} of RF (18.277%) continues to exceed that of GBRT (13.624%), with a smaller gap between them. Among neural network models, NN2 has the largest R^2_{oos} (15.326%), while NN4 has the smallest (12.053%). This implies that increasing hidden layers in neural networks does not enhance the R^2_{oos} of the model. Compared to mega-cap and large-cap cryptocurrencies, the difference in the predictabilities of neural network models becomes smaller, with R^2_{oos} ranging from 12% to 16%.

Achieving R^2_{oos} of 13.726%, the predictive performance of KNN outperforms GBRT, NN1, NN3, and NN4. The predictability of KNN experiences a notable decrease compared to mega-cap and large-cap cryptocurrencies. Notably, the R^2_{oos} of KNN is smaller than that of the full sample. The Diebold-Mariano tests for mid-cap cryptocurrencies are summarized in the Appendix. The findings are consistent with those obtained from the out-of-sample R^2 .

3.2 Cryptocurrency market vs. stock markets

We conduct a comparative analysis of the disparities in predictabilities of machine learning and linear models across the cryptocurrency and stock markets. To establish benchmarks for our evaluation, we leverage the findings of Gu et al. (2020) and Leippold et al. (2022) regarding the predictabilities of these models in the U.S. and Chinese stock markets, respectively.

When predicting return for the full sample of the U.S. and Chinese stock markets, OLS achieves R^2_{oos} of −3.46% and 0.81%, respectively. Notably, this value rises to 3.90% for the full sample of the cryptocurrency market. This suggests that OLS exhibits significantly superior predictive power within the cryptocurrency market compared to both stock markets, indicating a higher level of inefficiency within the former. In the cryptocurrency market and the U.S. and Chinese stock markets, the R^2_{oos} for PLS (ENet) are 3.90% (0.00%), 0.27% (0.11%), and 1.28% (1.42%), respectively. This underscores the significantly superior predictability of PLS within the cryptocurrency market. Conversely, ENet demonstrates no predictive power in the cryptocurrency market and weak (certain) predictive power in the U.S. (Chinese) stock market.

The R^2_{oos} for RF and GBRT in the cryptocurrency market are 25.27% and 24.91%, respectively. In the U.S. (Chinese) stock market, the corresponding values are 0.33% (2.44%) for RF and 0.34% (2.71%) for GBRT. In the cryptocurrency market, NN4 emerges as the optimal neural network model, achieving an R^2_{oos} of 18.70%. However, for the U.S. (Chinese) stock market, NN3 (NN5) stands out as the optimal neural network model, with an R^2_{oos} of 0.40% (2.58%). Thus, in the cryptocurrency (stock) market, tree (neural network) models demonstrate superior predictive power compared to neural network (tree) models, suggesting a divergence in model efficacy across asset classes. Tree-based models appear to excel in predicting cryptocurrency returns, while neural networks demonstrate superior performance in stock market prediction.

When predicting mega-cap cryptocurrencies, RF and GBRT achieve R^2_{oos} of 42.76% and 41.15%, respectively. This contrasts with their performance for U.S. large-cap stocks, where the R^2_{oos} fall to 0.63% (RF) and 0.52% (GBRT), and Chinese large-cap stocks, where both models exhibit negative R^2_{oos} (−0.04% and −0.38%), indicating no predictive power. These findings suggest that tree models demonstrate powerful and robust predictabilities for mega-cap cryptocurrencies, whereas their efficacies in predicting large-cap stocks are considerably weaker. The optimal neural network model for mega-cap cryptocurrencies, NN4, also demonstrates strong performance, achieving an R^2_{oos} of 36.09%. However, the optimal neural network models for U.S. and Chinese large-cap stocks, NN3, yield significantly lower R^2_{oos} (0.70% and 0.74%, respectively), indicating their limited predictive power in stock markets. This pattern reinforces the observations made with tree models, suggesting that neural networks also excel in predicting mega-cap cryptocurrencies but struggle with large-cap stocks. Further analysis for large-cap cryptocurrencies reveals similar trends. RF and GBRT attain R^2_{oos} of 27.37% and 21.48%, respectively. For the optimal neural network model, NN5, the R^2_{oos} is 19.27%. These findings suggest that both tree and neural network models exhibit strong predictive performance for large-cap cryptocurrencies compared to large-cap stocks.

In summary, our analysis reveals a substantial divergence in the predictabilities of machine learning and linear models across asset classes and capitalization segments. While demonstrating excellent predictive performance for cryptocurrencies, both tree-based models and neural networks show significantly weaker predictability for stocks in both the U.S. and Chinese markets.

3.3 Factor importance

We employ the SHAP method to investigate the importance of cryptocurrency factors in the full sample.⁴ Given the superior predictive performance of tree and neural network models, our emphasis is directed toward these particular model types. Furthermore, we conduct an analysis of the resemblances and variations in the factor importance across distinct tree or neural network models. The results of factor importance for mega-cap, large-cap, and mid-cap cryptocurrencies are provided in the Appendix. Figure 2 shows the importance of factors in tree and neural network models. Overall, the change rates of market value to realized value ratio (MRR), new addresses (NA), and active addresses (AA) are the three most influential factors for predicting weekly returns in the cryptocurrency market.

For tree models, MRR stands out as the most important factor, with a significantly higher level of importance compared to other factors. Following is NA, whose importance is approximately one-third and half that of MRR for RF and GBRT, respectively. Moreover, the change rates of up minus down (UMD), Google Trends cryptocurrency index (GTCX), crude oil price (WTI), US dollar index (USDIX), market return (RMRF), and active addresses ratio (AAR) exhibit lower degrees of importance. In RF, AA holds slightly larger importance than UMD, while GTCX is a little higher than WTI. Meanwhile, USDIX, RMRF, and AAR demonstrate nearly equivalent levels of importance. In GBRT, WTI surpasses UMD in importance, while USDIX and transactions volume (TV) exhibit comparable importance.

⁴ SHAP is a game theoretic approach to explain the output of any machine learning model. The details of the SHAP method are provided in the Appendix.

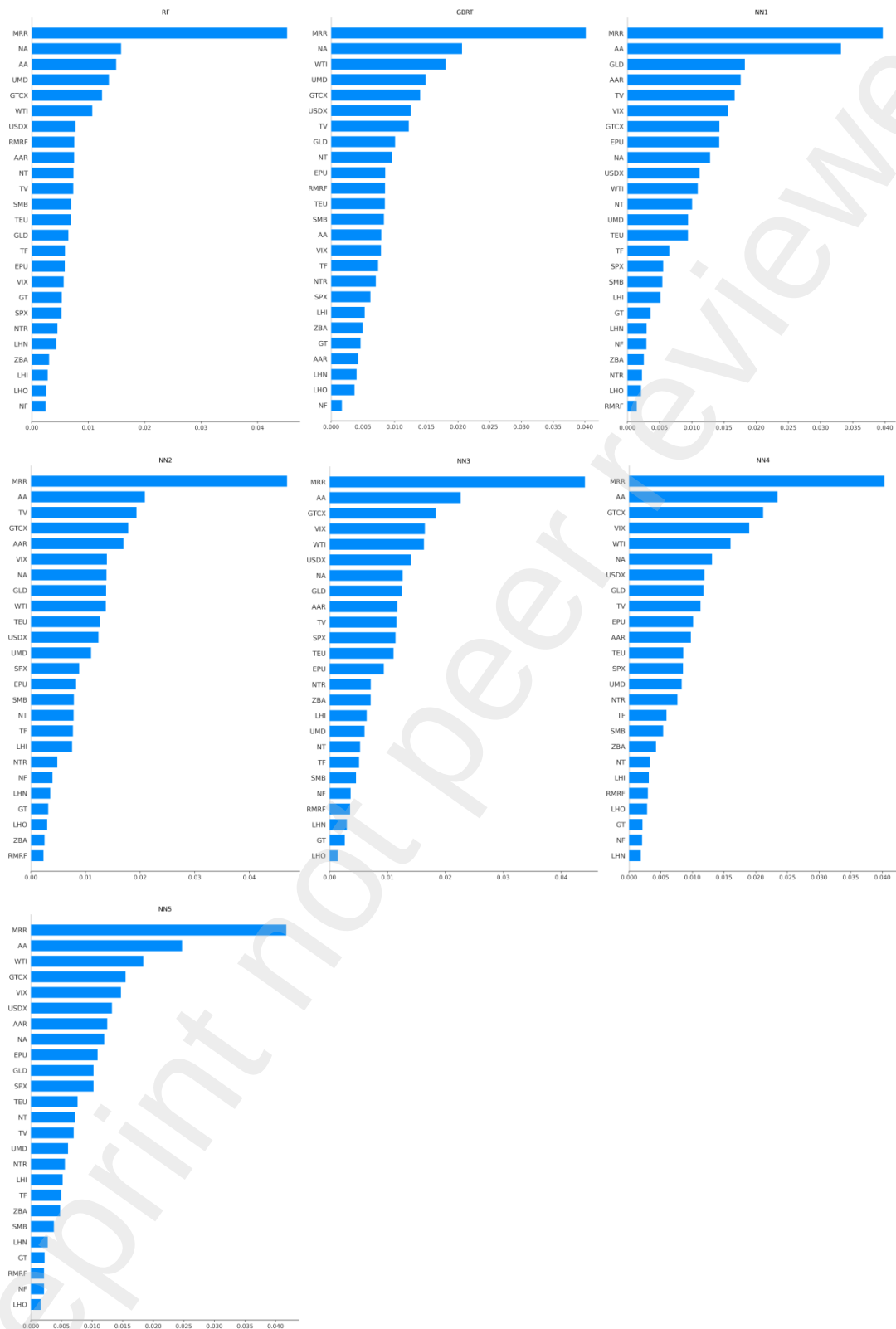


Figure 2. The importance of cryptocurrency factors in each tree and neural network model.

Notes: The horizontal axis denotes the average absolute value of the SHAP values, serving as a metric for factor importance. The larger the value, the more important the factor. The vertical axis represents the factors' names, arranged in descending order based on their importance levels.

MRR is also the most important factor in neural network models, and its importance is markedly higher than others, except for NN1. In contrast to tree models, neural network models exhibit a pronounced preference for AA, especially for NN1. Notably, the importance of NA is significantly decreased in neural network models. Given the superior performance of NN4 compared to other neural network models, we mainly focus on discussing the importance of factors in this model. In NN4, AA takes precedence as the second crucial factor, with its importance being nearly half of that assigned to MRR. Some factors, such as the change rates of GTCX, volatility index (VIX), WTI, USDX, NA, and gold price (GLD), also obtain considerable priority in NN4. Furthermore, VIX and WTI have comparable importance, while GTCX shows lower importance to AA.

To provide a comprehensive overview, Figure 3 shows the overall importance of all factors in both tree and neural network models based on the full sample.

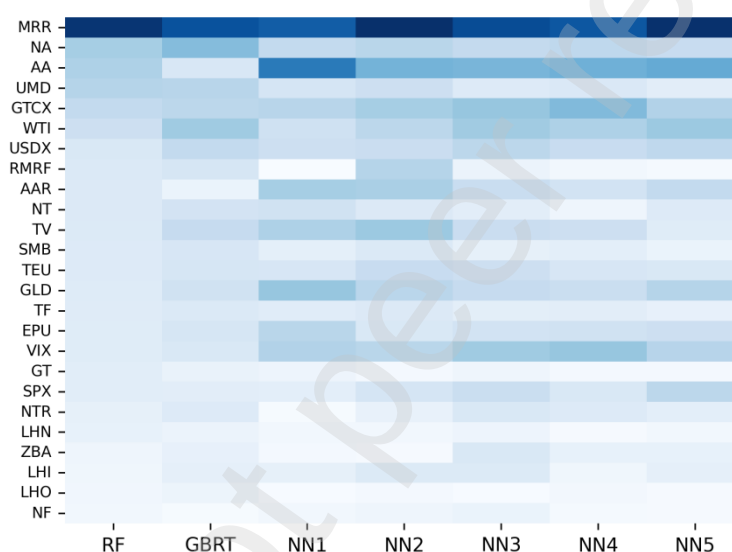


Figure 3. The overall importance of cryptocurrency factors for tree and neural network models.

Notes: This figure illustrates the ranking of all factors based on their overall contribution to the model. Factors on the vertical axis are ordered based on the average absolute value of the SHAP values, with the most influential characteristics on the top and the least influential on the bottom. Columns correspond to the individual models, and the color gradients within each column indicate the most influential (dark blue) to the least influential (white) factors.

3.4 Factor effects on model output

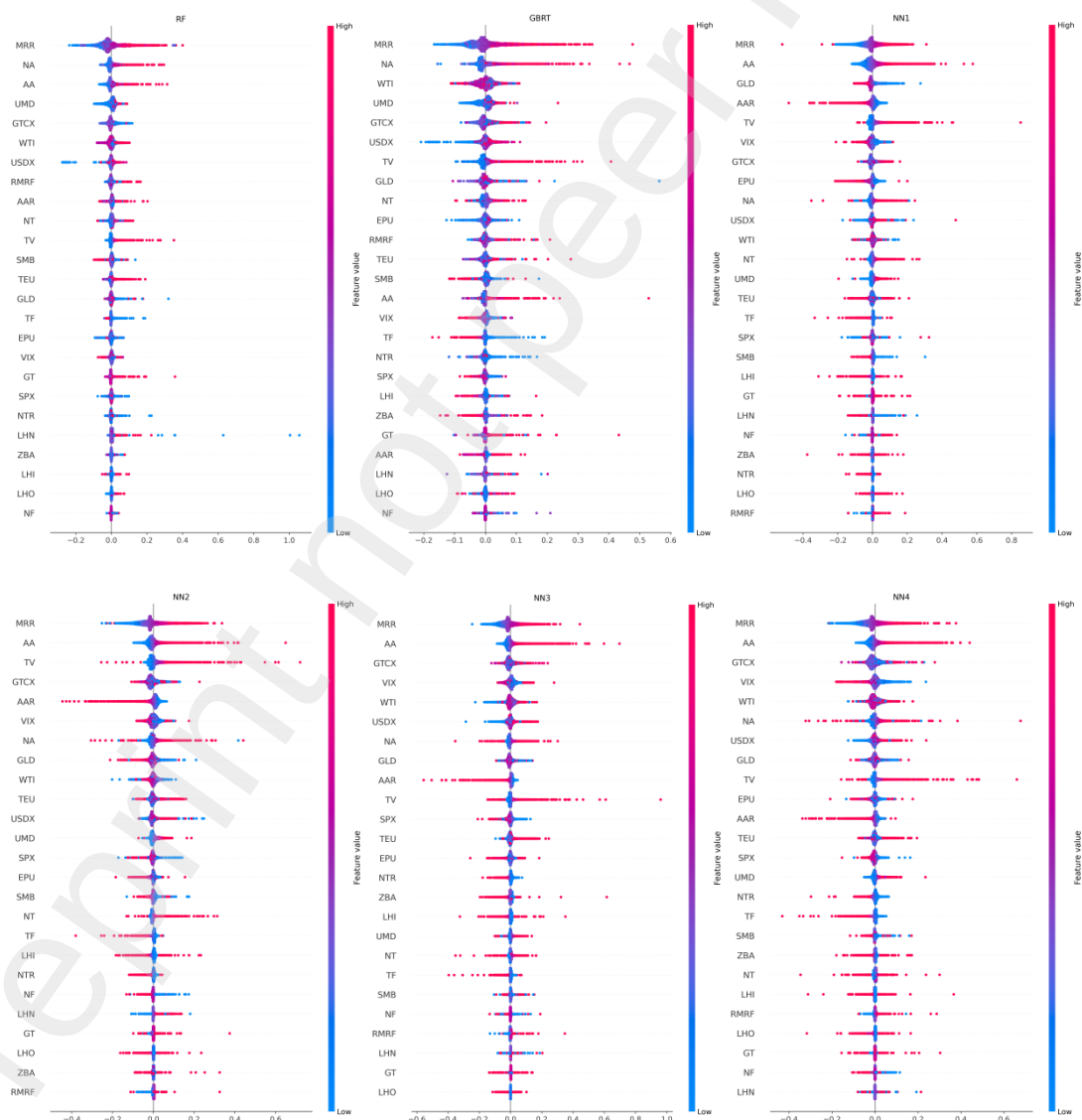
We also apply the SHAP method to understand the decision-making processes of tree and neural network models in the full sample. This methodology enables the examination of the impact of cryptocurrency factors on model output, as illustrated in Figure 4. The corresponding results for mega-cap, large-cap, and mid-cap cryptocurrencies are provided in the Appendix.

As the three most important factors, MRR, NA, and AA positively influence the output of all models. The magnitudes of their values correlate positively with SHAP values, consequently increasing the predicted returns. Interestingly, the effect of MRR and NA are notably more significant than other factors in tree models, while the impact of AA slightly exceeds MRR in neural network models.

In the case of tree models, WTI, USDX, and TV also positively contribute to the output, whereas GLD and total flow (TF) have a negative impact. The impacts of NA and AA are almost identical in RF, and those of NA and TV are nearly the same in GBRT. Notably, the influence of AA in RF is slightly greater than that in GBRT, while the impact of NA in GBRT is more substantial than that in RF. Furthermore, the adverse effect of TF in GBRT is also more significant than that in RF.

In the context of NN4, factors such as USDX and UMD also have a positive influence on model output, although their impact is notably smaller compared to MRR and AA. However, VIX, network value to transaction ratio (NTR), and TF negatively contribute to model output, with VIX exhibiting a significantly larger influence. It is noteworthy that the impacts of GTCX, WTI, NA, GLD, and TV on the output are unclear.

Figure 5 visualizes the contribution of cryptocurrency factors to model prediction, utilizing RF, GBRT, and NN4 as representative examples. These instances offer valuable insights into how each factor drives the forecast toward its final value.



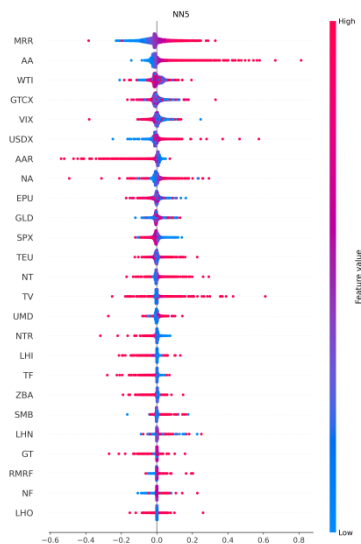


Figure 4. The impact of cryptocurrency factors on tree and neural network models.

Notes: The horizontal axis signifies the SHAP value, reflecting the impact of factors on the model output. A larger SHAP value denotes a more substantial influence on the model output. The vertical axis represents the factors' names, arranged in descending order based on their relative importance. Additionally, the color gradient indicates the factor values; a redder hue corresponds to a higher factor value, while a bluer shade indicates a lower factor value.

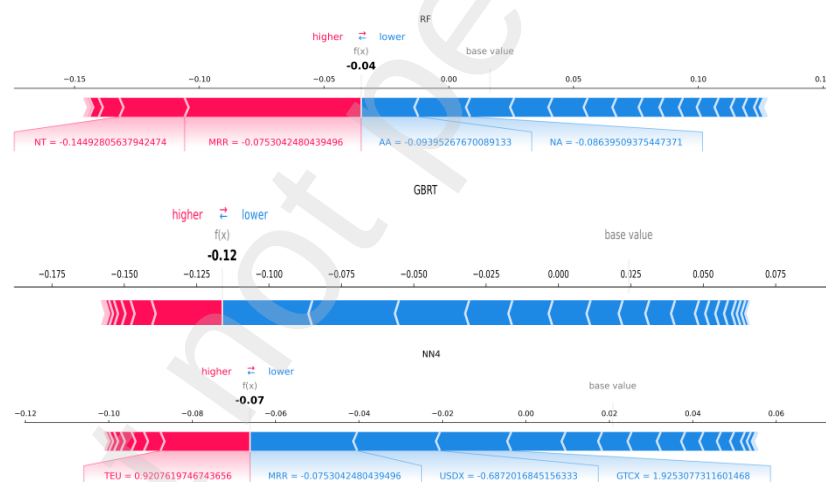


Figure 5. Examples of cryptocurrency factors' contribution to machine learning models.

Notes: The horizontal axis denotes the factor, while the vertical axis represents the cumulative impact of the factor originating from the base value (the average prediction of the model). Arrows are employed to signify the contribution of individual factors, with arrow length and color indicating the magnitude and direction of the factor's influence, respectively. A red hue denotes a positive impact, pushing forecasts higher, whereas blue signifies a negative impact, driving forecasts lower. Black bold indicates the final predicted value for a specific case.

4. Portfolio analysis

So far, our evaluation of predictive performance has been grounded in statistical comparisons, utilizing the out-of-sample predictive R^2 and Diebold-Mariano test. Next, we analyze whether this predictability can be implemented at portfolio level by assessing the out-of-sample performance of

value-weighted portfolios and tracking the evolution of their cumulative returns over time.⁵ While we also consider equal-weighted portfolios, it is noteworthy that they exhibit inferior performance compared to the value-weighted portfolios. Thus, the detailed results of the equal-weighted portfolios are presented in the Appendix. The benefit of investigating predictability at the portfolio level lies in the ability to evaluate the economic impact of each method through its contribution to risk-adjusted portfolio return performance.

We consider two types of portfolios. The first type is long-short portfolios, which are constructed by following the schemes of Gu et al. (2020) who evaluate machine learning portfolios in the U.S. stock market.⁶ Specifically, we employ each method at the end of each week to predict the out-of-sample returns of cryptocurrencies for the next week. The cryptocurrencies are then divided into deciles based on the predicted returns, and both value-weighted and equal-weighted portfolios are constructed. Thus, a zero-net-investment portfolio is built by buying the cryptocurrencies with the highest expected returns (decile 10) and selling those with the lowest (decile 1). It is noteworthy that implementing long-short portfolios is difficult in the current cryptocurrency market due to the absence of a short-selling mechanism, although they are a useful tool for assessing the portfolio-level performance of machine learning methods. Hence, we also consider long-only portfolios, which only buy the cryptocurrencies in the top decile.

Table 5 reports the out-of-sample performance for value-weighted long-short and long-only portfolios, suggesting that machine learning portfolios dominate linear ones. Overall, machine learning techniques, especially PLS and NNs, are beneficial for portfolio-level predictions. In long-short portfolios, linear portfolios exhibit inferior performance compared to their machine-learning counterparts. The OLS-3 portfolio stands out with the smallest mean and the largest maximum drawdown, indicating that OLS utilizing only three factors is unfavorable for predicting portfolio returns. The NN5-based portfolio records the largest mean, followed by the NN4-based portfolio. Thus, the deep neural network portfolio is the most proficient when evaluating investment returns. Notably, the Sharpe ratios of neural network portfolios are not the highest, whereas the PLS-based portfolio surprisingly achieves the largest Sharpe ratio and the smallest standard deviation. Furthermore, the NN1-based portfolio exhibits the smallest maximum drawdown, trailed by the RF-based portfolio.

In the context of long-only portfolios, the neural network portfolio demonstrates the best return and investment efficiency. Specifically, portfolios based on NN4 and NN3 reach the largest mean and Sharpe ratio, respectively. NN5-based portfolio secures the second-largest mean. In addition, the RF-based portfolio exhibits the second largest Sharpe ratio, while the GBRT-based portfolio attains the smallest maximum drawdown. The PLS-based portfolio maintains its low variation by having the smallest standard deviation.

⁵ We construct the value-weighted portfolios by weighting the market capitalizations of the cryptocurrencies in each portfolio.

⁶ Leippold et al. (2022) also adopt the methodologies outlined by Gu et al. (2020) to construct machine learning portfolios in the Chinese stock market.

Table 5

The out-of-sample performance for value-weighted portfolios on the full sample.

	Mean	Std	Sharpe	Skewness	Kurtosis	MaxDD
Long-short						
OLS	11.375	17.528	63.698	0.506	−0.572	90.064
OLS-3	6.829	19.395	34.012	−0.193	2.571	97.327
PLS	14.393	17.372	81.258	0.428	−0.570	88.985
LASSO	13.132	21.737	59.314	4.152	25.542	89.378
ENet	13.616	21.298	62.139	4.173	25.801	89.463
Ridge	19.847	25.449	59.821	1.451	2.875	89.763
RF	26.369	34.395	75.566	1.733	3.430	88.552
GBRT	28.531	36.297	77.406	1.722	3.270	88.636
KNN	20.972	29.398	70.853	1.654	3.806	89.642
NN1	24.528	34.426	70.842	1.820	3.627	87.879
NN2	24.394	36.755	65.486	2.640	10.113	89.546
NN3	26.781	38.573	68.924	2.293	6.458	89.243
NN4	28.249	42.124	66.450	2.328	7.225	89.431
NN5	29.883	43.362	67.519	2.314	6.623	89.352
Long-only						
OLS	9.372	16.472	55.897	0.251	3.489	87.357
OLS-3	5.484	28.356	18.340	1.298	8.416	95.435
PLS	11.241	16.247	68.188	0.922	2.696	89.472
LASSO	10.365	20.343	49.951	0.197	6.334	98.455
ENet	10.385	20.462	49.753	0.545	6.313	95.378
Ridge	14.836	21.473	68.235	−0.190	5.604	90.425
RF	22.242	31.352	69.043	1.607	3.676	82.782
GBRT	24.081	34.546	68.782	2.419	7.405	79.386
KNN	17.384	25.463	67.272	0.355	4.040	90.535
NN1	21.537	35.354	59.918	1.844	3.937	84.536
NN2	22.524	31.978	65.890	1.237	4.064	86.643
NN3	23.390	33.275	69.989	0.828	2.253	89.264
NN4	25.138	38.472	64.341	1.663	3.070	85.543
NN5	24.347	38.573	61.045	1.540	2.643	86.744

Notes: This table reports the out-of-sample performance measures for both long-short and long-only portfolios constructed by all machine learning models. The portfolios are value-weighted and based on the full sample. All measures are derived from the out-of-sample returns from January 2014 to March 2023. “Mean” represents the average predicted weekly return (%). “Std” denotes the standard deviation of the predicted weekly returns (%). “Sharpe” represents the annualized Sharpe ratio (%). “MaxDD” denotes the maximum drawdown (%) of the portfolio.

The dynamic evolution of cumulative returns for both long-short and long-only value-weighted portfolios is shown in Figure 6. Note that we select the NN5-based (NN4-based) portfolio to represent neural network portfolios under the long-short (long-only) strategy, given its best

performance among neural network portfolios for each respective strategy. The dynamics of cumulative returns for long-short and long-only portfolios based on NN1-NN5 will be discussed later.

The cumulative return performance shows that long-short portfolios can be divided into 4 tiers. The first tier contains portfolios based on NN5, GBRT, and RF, demonstrating significantly higher cumulative returns than other portfolios. Within this tier, the NN5-based portfolio achieves the largest cumulative return, followed by the GBRT-based portfolio. The second tier comprises portfolios based on KNN and Ridge, with the former's cumulative return notably exceeding the latter's. Notably, there is a substantial gap between the first and second tiers. Portfolios based on PLS, OLS, LASSO, and ENet are composed of the third tier, with a certain gap between the second and third tiers. Among these portfolios, the PLS-based portfolio attains the largest cumulative return, trailed by the OLS-based portfolio. The cumulative returns of portfolios based on LASSO and ENet are nearly the same. Finally, the OLS-3 portfolio forms the fourth tier and exhibits the smallest cumulative return. The gap between the last two tiers has increased markedly since the third quarter of 2021.

On the other hand, long-only portfolios can be classified into 5 tiers. The first tier includes portfolios based on NN4, GBRT, and RF, whose cumulative returns are notably larger than those of other portfolios. Notably, the NN4-based portfolio attains the largest cumulative return, followed by the GBRT-based portfolio. In addition, the RF-based portfolio is comparable to the GBRT-based portfolio. The KNN-based portfolio forms the second tier, exhibiting an obvious gap from the first tier, especially after the fourth quarter of 2021. The third tier comprises portfolios based on Ridge and OLS, with the former having higher cumulative returns. A substantial gap emerges between the second and third tiers, particularly pronounced after November 2021. The PLS-based portfolio builds the fourth tier, maintaining a distinct separation from the OLS-based portfolio. Finally, portfolios based on LASSO, ENet, and OLS-3 are the fifth tier, sharing the lowest cumulative returns with minimal variations among them.

It is noteworthy that regardless of long-short or long-only, machine learning portfolios exhibited robust performance during the cryptocurrency market crash spanning from November 2021 to June 2022. Interestingly, the shock caused by the market crash did not lead to a universal decline across all portfolios. Machine learning portfolios, especially those based on neural networks and trees, demonstrated exceptional resilience during this period. Their cumulative returns experienced a substantial upswing, contrasting the prevailing market trend. This resilience underscores the potential advantages and effectiveness of both neural network and tree models in adapting to and thriving in volatile market scenarios.

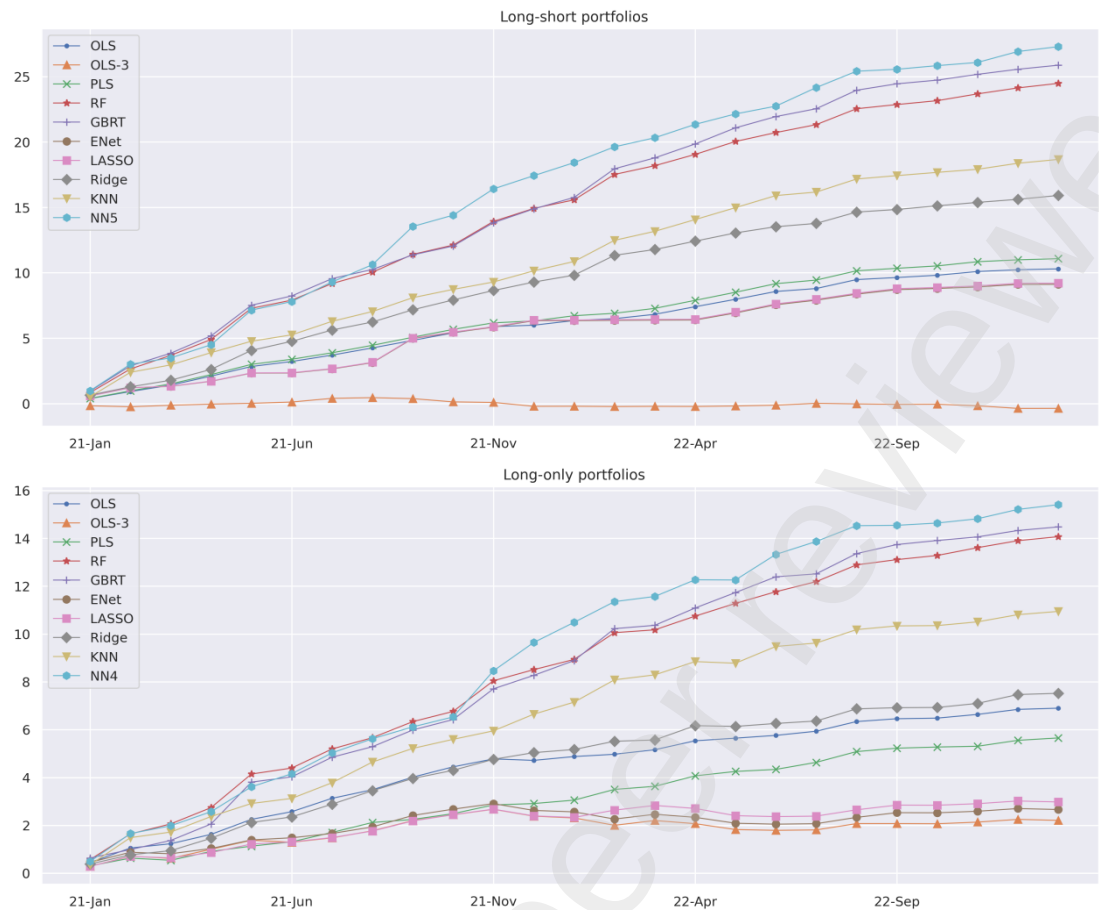


Figure 6. Cumulative returns of value-weighted portfolios based on the full sample.

Figure 7 illustrates the dynamics of cumulative returns for five neural network portfolios. Long-short neural network portfolios can be split into 3 tiers. Portfolios based on NN4 and NN5 compose the first tier, boasting significantly higher cumulative returns. However, the former exhibits a marginally higher cumulative return. The second tier includes portfolios based on NN1 and NN3, demonstrating comparable cumulative returns. The NN1-based portfolio shows a slightly higher cumulative return. The NN2-based portfolio forms the third tier, with the smallest cumulative return. Notably, significant gaps separate each tier. On the other hand, short-only neural network portfolios also exhibit a three-tier classification. The first tier contains only the NN4-based portfolio, while the second maintains portfolios based on NN1 and NN3. It is noteworthy that the NN5-based portfolio experiences a significant decline, placing it in the third tier alongside the NN2-based portfolio. The former shows a slightly higher cumulative return within this tier. During the market crash, neural network portfolios performed well, with their cumulative returns rising against the trend. Long-short portfolios displayed a steady upward trajectory, while long-only portfolios experienced small fluctuations. The cumulative returns of neural network portfolios have demonstrated a slow increase since August 2022.

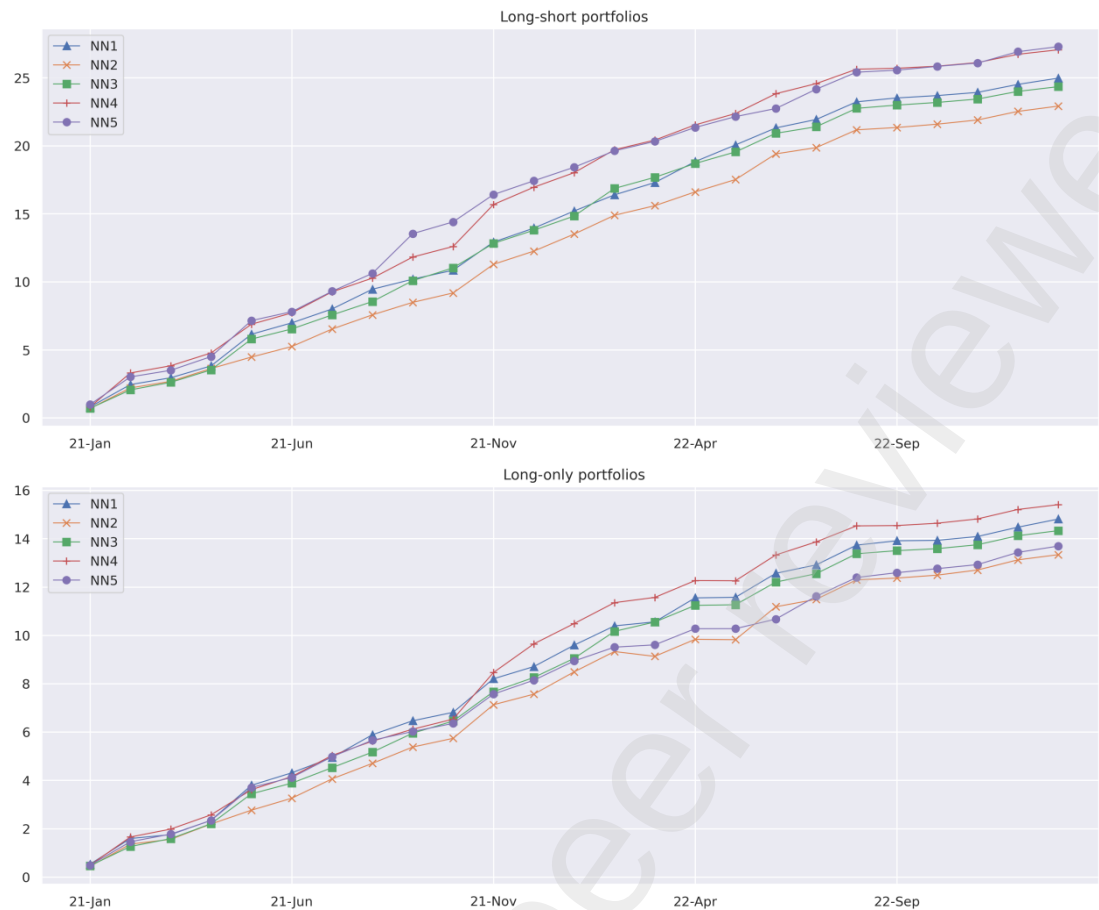


Figure 7. Cumulative returns of neural network portfolios (value-weighted) based on the full sample.

5. Conclusion

In this study, we construct a comprehensive set comprising 8 macroeconomic and 17 characteristic factors to predict cryptocurrency returns. Leveraging this extensive set of factors, we systematically explore the predictive power of 12 distinct machine learning methods in conjunction with 2 linear methods in the cryptocurrency market. Furthermore, we also examine which factors matter the most in specific models and assess how factors influence the output of the models. Finally, we investigate the predictive performance of long-short and long-only portfolios constructed by these diverse methodologies. Our study allows for an exhaustive evaluation of the predictabilities and practical implications of machine learning techniques within the cryptocurrency domain.

We find that both tree and neural network models exhibit exceptional predictabilities, with the random forest model demonstrating the best predictive performance. Notably, most models achieve their best predictive performance when applied to mega-cap cryptocurrencies, whereas the worst when applied to mid-cap ones. Surprisingly, all models show a slightly superior performance on the full sample compared to their performance on mid-cap cryptocurrencies. Our findings also show that the change rates of market value to realized value ratio, new addresses, and active addresses are the most important factors for tree and neural network models. By exploring the decision-making processes of black-box models, these factors demonstrate a significantly positive impact on model outputs. This examination not only underscores the importance of specific factors but also sheds light on the nuanced mechanism influencing the predictabilities of machine learning models in the cryptocurrency market.

Our portfolio analysis reveals a clear dominance of long-short and long-only machine learning portfolios over their linear counterparts. When considering the out-of-sample performance metrics, portfolios based on PLS and NNs are superior to other machine learning portfolios. Moreover, the long-short and long-only portfolios based on NN4, NN5, RF, and GBRT demonstrate significantly higher cumulative returns compared to other portfolios. Notably, cumulative returns of machine learning portfolios, especially those derived from neural networks and tree models, experienced a substantial increase during the cryptocurrency market crash. This resilience and positive performance during adverse market conditions underscore the robustness and adaptability of machine learning portfolios.

In summary, machine learning methods can be successfully employed in the cryptocurrency market. The SHAP method allows for a transparent illustration of the internal mechanisms of these black-box models. We provide valuable insights for investors, traders, and financial analysts, offering guidance for the pricing of cryptocurrency assets.

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Internet Appendix

A Methodology

A.1 Ordinary least squares (OLS) model

The OLS model is a simple linear model and assumes that the conditional expectation can be approximated by a linear function of predictor variables and parameter vectors, that is

$$g(z_{i,t};\theta) = z'_{i,t}\theta,$$

where $g(\cdot)$ represents the conditional expectation, $z_{i,t}$ denotes predictor variables, and θ represents parameter vector. The model uses simple regression and does not allow for nonlinear effects or interactions between predictor variables. The standard least squares method is used to estimate the above model, and the objective function is

$$\mathcal{L}(\theta) = \frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T [(r_{i,t+1} - g(z_{i,t};\theta))]^2,$$

where $\mathcal{L}(\theta)$ represents the loss function. The above objective function is called l_2 , which can provide analytical estimates, thus avoiding complex optimization and calculations. Finally, minimizing $\mathcal{L}(\theta)$ yields the combined OLS estimator.

A.2 Partial least squares (PLS) model

The PLS model combines the features of principal component analysis and multivariate regression and is suitable for predicting a set of dependent variables (prediction targets) from a set of high-dimensional independent variables (predictors). The model utilizes the covariance of predictors and prediction targets for dimensionality reduction. For each predictor j , its univariate return prediction coefficient φ_j is estimated through OLS regression, which reflects the partial sensitivity of returns to each predictor j . All predictors are then averaged to obtain a single composite component weighted proportionally to the coefficient φ_j placing the greatest (minimum) weight on the strongest (weakest) univariate predictor. In this way, PLS performs dimensionality reduction considering the final prediction goal. To form multiple prediction components, the prediction target and all predictors are orthogonalized to the previously constructed components, and the above process is iterated on the orthogonalized data set until the required number of PLS components is reached.

Rewrite the linear regression model, $r_{i,t+1} = z'_{i,t}\theta + \varepsilon_{i,t+1}$, in matrix form

$$R = Z\theta + E,$$

where R is an $NT \times 1$ dimensional vector of returns $r_{i,t+1}$, Z is an $NT \times P$ dimensional matrix of stacked predictors $z_{i,t}$, and E is the error $\varepsilon_{i,t+1}$ is an $NT \times 1$ dimensional vector. The PLS model uses a general method to reduce the dimensionality, compressing the P dimensional predictor set into K (much smaller than P) linear combinations of predictors. Thus, the above model can be written as

$$R = (Z\Omega_K)\theta_K + E,$$

where Ω_K is a $P \times K$ dimensional matrix, and each column is $w_1, w_2, \dots, w_K$. Each w_j is the set of linear combination weights used to create the j th predictive component. Thus, $Z\Omega_K$ is a

reduced-dimensional form of the original set of predictors, and the prediction coefficients θ_K become a $K \times 1$ dimensional vector.

The goal of the PLS model is to find K linear combinations regarding Z that have the greatest predictive correlation with the prediction target. The weights used to construct the j th partial least squares component solve the following optimization problem

$$w_j = \max_w \text{Cov}^2(R, Zw), \quad \text{s.t. } w'w = 1, \text{Cov}(Zw, Zw_l) = 0, l = 1, 2, \dots, j-1.$$

Indeed, the PLS model finds components with more effective return predictability by sacrificing the accuracy with which $Z\Omega_K$ approximates Z . Given the solution for Ω_K , an estimate of θ_K can be obtained by OLS regression of R on $Z\Omega_K$. We adaptively optimize hyperparameter K using samples on the validation set.

A.3 Elastic net (ENet) model

Compared to the simple linear model, the penalized linear model adds a penalty term to the loss function

$$\mathcal{L}(\theta; \cdot) = \mathcal{L}(\theta) + \phi(\theta; \cdot),$$

where $\mathcal{L}(\theta)$ is the original loss function, and $\phi(\cdot)$ is the new penalty function added. There are many options for this penalty function. When $\phi(\cdot)$ is in the following form

$$\phi(\theta; \lambda, \rho) = \lambda(1 - \rho) \sum_{i=1}^N |\theta_i| + \frac{1}{2} \lambda \rho \sum_{i=1}^N \theta_i^2,$$

where λ and ρ are penalty hyperparameters and are non-negative. The $\mathcal{L}(\theta; \cdot)$ model now is called the ENet model. We adaptively optimize the above hyperparameters using samples on the validation set. We set $\alpha = 0.5$ and l_1 ratio as 0.5.

A.4 Least absolute shrinkage and selection operator (LASSO) model

For the penalty function $\phi(\theta; \lambda, \rho)$ in the ENet model, when the hyperparameter $\rho = 0$, this function becomes

$$\phi(\theta; \lambda) = \lambda \sum_{i=1}^N |\theta_i|.$$

The above penalty function is called the l_1 penalty parameter, which sums the absolute values. Now, the $\mathcal{L}(\theta; \cdot)$ model is called the LASSO model. The geometry of the LASSO model sets the coefficients of a subset of covariates to zero, i.e., the model imposes sparsity on the specification. Therefore, the LASSO model can be regarded as a variable selection method. We adaptively optimize hyperparameter λ using samples on the validation set. We set $\alpha = 1$.

A.5 Ridge model

For the penalty function $\phi(\theta; \lambda, \rho)$ in the ENet model, when the hyperparameter $\rho = 1$, this function becomes

$$\phi(\theta; \lambda) = \frac{1}{2} \lambda \sum_{i=1}^N \theta_i^2.$$

The above penalty function sums the squares, which is essentially the objective function l_2 in the

OLS model. Now, the $L(\theta; \cdot)$ model is called a Ridge model. The l_2 penalty parameter brings all coefficient estimates close to zero but does not impose exact zeros anywhere. Therefore, the ridge model is a shrinkage method that helps prevent the size of the coefficients from becoming too large. We adaptively optimize hyperparameter λ using samples on the validation set. We set $\alpha = 0.8$.

A.6 Random forest (RF) model

The RF model is an ensemble method that combines the predictions of many different trees. According to Breiman (2001), the RF model is a variant of a more general process, also known as bootstrap aggregation or bagging. The baseline tree bagging process draws different bootstrap samples of the data, fits a separate regression tree for each sample, and then averages their predictions. Trees for individual bootstrap samples tend to be deep and overfitted, which makes their predictions inefficient. Averaging multiple predictions can reduce this variation, thus stabilizing the tree's predictive performance.

Using different bagging methods, the RF model reduces the correlation between trees in different bootstrap samples. For example, if the size of a cryptocurrency is the dominant predictor of returns in the data, then most bagging trees will perform low-level splits on size, resulting in significant correlations between their final predictions. The forest method uses a dropout method to de-correlate the tree, which only considers a randomly drawn subset of predictors for splitting at each potential branch. In the above example, at least a few early branches of the tree will be divided based on characteristics other than the cryptocurrency scale. Relative to standard bagging, this reduces the average correlation between predictors, further reducing variance. Furthermore, the tree depth (L) and the number of bootstrap samples (B) are hyperparameters optimized by the validation sample.

The algorithm of the RF model is as follows: a tree is generated by a classification and regression tree algorithm. Generate bootstrap samples $\{(z_{i,t}, r_{i,t+1}), (i, t) \in \text{Bootstrap}(b)\}$ from the original data set. At each step of segmentation, only random subsamples ($\sqrt{P}$) of all features are used. For the b th tree ($b \in [1, B]$), it can be written as

$$\hat{g}_b(z_{i,t}; \hat{\theta}_b, L) = \sum_{k=1}^{2^L} \theta_b^{(k)} \mathbf{1}\{z_{i,t} \in C_k(L)\},$$

where $z_{i,t}$ represents the predictor, $C_k(L)$ denotes one of K partitions of the data, each partition is the product of at most L indicator functions of the predictor, θ_k represents the constant related to partition k , which is defined as the sample mean of the results within a partition. Finally, the average of the output of all B trees is the final output of the RF model, i.e.,

$$\hat{g}(z_{i,t}; L, B) = \frac{1}{B} \sum_{b=1}^B \hat{g}_b(z_{i,t}; \hat{\theta}_b, L).$$

There are fewer hyperparameters in the RF model, and the model performance is robust. Thus, only one RF model is used in this paper. We set depth $L = 1 \sim 3$, trees $B = 100$, and features $n = 50$.

A.7 Gradient boosted regression trees (GBRT) model

The GBRT model combines boosting and decision trees to create an integrated model that excels at capturing complex patterns and relationships in data. The model can handle complex prediction tasks and produce highly accurate results. The GBRT model iteratively adds weak decision trees to

a growing ensemble to build a predictive model. Each subsequent tree is constructed to correct the errors produced by the previous tree, and the resulting ensemble collectively improves prediction accuracy.

Gradient descent optimization is a key component of the GBRT model, which determines the direction and magnitude of subsequent improvement of each tree. This algorithm calculates the negative gradient of the loss function with respect to the predicted value. This gradient guides the creation of new trees and minimizes errors caused by previous models. By adjusting the model's predictions toward the steepest descent, the model ensures that subsequent trees effectively correct ensemble errors.

The GBRT model introduces a learning rate to control the contribution of each subsequent tree to the ensemble. A lower learning rate means each tree has less impact on the final prediction, resulting in a relatively conservative model. Conversely, higher learning rates allow each tree to have a more substantial impact, which can lead to overfitting.

The algorithm of the GBRT model is as follows. First fit a shallow tree, for example, the depth of the tree is $L = 1$. This shallow tree will have a large amount of bias in the training samples and is a weak predictor. Then, a second shallow tree (with the same depth L) is used to fit the prediction error of the first tree. The predictions from the two trees are added together to get an overall prediction about the outcome, but the prediction components of the second tree are scaled down by a factor $v \in (0,1)$, thus preventing model overfitting errors. At each new step b , a shallow tree is fitted to the error of the model with $b - 1$ trees, and its error prediction is added to the total with shrinkage weight v . This process is iterated until there are a total of B trees in the set. Therefore, the final output is a shallow tree additive model with 3 hyperparameters (L, v, B) . We employ samples on the validation set to optimize these hyperparameters adaptively. In addition, we only utilize one GBRT model since this model has fewer hyperparameters and is robust. We set depth $L = 1 \sim 3$, trees $B = 100$, features $n = 50$, and learning rate $\alpha = 0.01$.

A.8 K-nearest neighbor (KNN) model

The KNN model is a non-parametric supervised learning algorithm that classifies objects based on the category of their nearest neighbor. Compared to the linear regression model, this model can deal with the nonlinear relationship between the independent and dependent variables. The computational cost of the KNN model is high, especially for large data sets or high-dimensional feature spaces.

The algorithm of the KNN model is as follows. The KNN model first finds the k data points most similar to a new one. The similarity between two data points is usually measured by, for example, Euclidean distance or Manhattan distance. Once the k most similar data points are found, the model utilizes the labels of these data points to predict the labels of new data points. The label of a new data point is the most common label among the k most similar data points. In addition, the model is sensitive to the choice of the value of k . We set the number of neighbors as 10.

A.9 Neural network (NN) model

We use i to denote the layer in the NN model. Assume there are $n + 2$ layers in total, where n is an integer. When $i = 0$, layer i represents the input layer. When $1 \leq i \leq n$, layer i represents the hidden layer. When $i = n + 1$, layer i represents the output layer. Assume that the input layer ($i = 0$) has k_0 input variables (features), the hidden layer i ($1 \leq i \leq n$) has k_i neurons, and the

output layer ($i = n + 1$) has k_{n+1} output variables (targets). We treat features and targets as special neurons and then assume that on the i th layer, the value of its j th neuron is $z_{i,j}$, where $0 \leq i \leq n + 1$ and $1 \leq j \leq k_i$. Then, $z_{0,j} = x_j$ represents the value of the j th feature of the input layer, $z_{n+1,j} = y_j$ represents the estimated value of the j th target of the output layer; for $1 \leq i \leq n + 1$ and $1 \leq j \leq k_i$, the calculation formula of $z_{i,j}$ is

$$z_{i,j} = f_i \left(b_{i,j} + \sum_{l=1}^{k_{i-1}} w_{i-1,l,j} z_{i-1,l} \right),$$

where f_i is the activation function, which is used to calculate the value of the neuron on the i th layer; $w_{i,l,j}$ is the weight, which means that when calculating the value of the j th neuron on the $i + 1$ th layer, it is assigned to the weight of the value of the l th neuron on the i th layer; $b_{i,j}$ is the bias, which is a constant and is used to calculate the value of the j th neuron on the i th layer. Like the normalization process in initialization, we set all initial weights to small positive constants.

For the network structure of the NN model, we consider various structures with up to 5 hidden layers and select the number of neurons per layer according to the geometric pyramid rule (Masters, 1993). The shallowest network has only 1 hidden layer with 32 neurons, denoted by NN1. Next, NN2 has 2 hidden layers with 32 and 16 neurons, respectively. NN3 has 3 hidden layers with 32, 16, and 8 neurons, respectively. NN4 has 4 hidden layers with 32, 16, 8, and 4 neurons, respectively. Finally, NN5 represents the deepest neural network with 5 hidden layers with 32, 16, 8, 4, and 2 neurons, respectively. All structures in a neural network model are fully connected, thus each unit receives input from all units in the underlying layer. By comparing the performance of NN1 to NN5, we can infer the trade-off of network depth.

We use the AMSGrad algorithm in the NN models. The algorithm proposed by Reddi et al. (2019) maintains the maximum value of the second-order moment vector and uses it to bias the gradient to update the parameters. For the hyperparameters, we set $\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\varepsilon = 10^{-8}$. In addition, the batch size is set to 16, i.e., each iteration randomly uses 16 samples in the training set.

We have considered 5 commonly used activation functions (Sigmoid, Tanh, ReLU, Leaky ReLU, and Exponential LU functions) and find that the Sigmoid function leads to the best result. Therefore, we use this activation function in each layer, which is given by

$$f_i(x) = \frac{1}{1 + e^{-x}}.$$

We employ the early stopping to prevent overfitting of the NN model during the training process. As the model's error on the training set continues to decrease, the training will stop if the model does not reduce the error on the validation set for 20 epochs. The model will terminate training after 1000 epochs if the early stopping is not triggered.

A.10 Diebold and Mariano test

For pairwise comparisons of methods, we use the Diebold and Mariano (1995) test for differences in out-of-sample predictive accuracy between two models. Specially, we define the test statistic below to test the forecast performance of method 1 and 2

$$DM_{12} = \frac{\bar{d}_{12}}{\hat{\sigma}_{\bar{d}_{12}}},$$

$$\bar{d}_{12,t+1} = \frac{1}{n_{3,t+1}} \sum_{i=1}^{n_{3,t+1}} ((\hat{e}_{i,t+1}^{(1)})^2 - (\hat{e}_{i,t+1}^{(2)})^2),$$

where $\hat{e}_{i,t+1}^{(1)}$ and $\hat{e}_{i,t+1}^{(2)}$ represent the prediction error for cryptocurrency i at time t utilizing each method, and $n_{3,t+1}$ is the number of cryptocurrencies in the testing sample (week $t + 1$). Then $\bar{d}_{12}$ and $\hat{\sigma}_{\bar{d}_{12}}$ denote the mean and standard error of $d_{12,t}$ over the testing sample. If the absolute value of the test statistic is greater than the significance level of 5%, we reject the null hypothesis that there is no difference in forecast accuracy between the two methods.

A.11 Shapley additive explanations (SHAP)

The SHAP method is a powerful tool in machine learning for explaining model predictions by attributing them to individual input features.⁷ It draws inspiration from cooperative game theory, particularly Shapley values introduced by Lloyd Shapley (1953). SHAP values offer a principled and fair way to distribute the contribution of each feature to the model's prediction, providing insights into the importance and impact of different input variables.

SHAP values leverage the cooperative game theory concept of Shapley values, which determine a fair distribution of payoffs among players in a coalition. In machine learning, the “players” are the input features of a model, and the “payoff” is the model's prediction for a specific instance. By calculating the Shapley values for each feature, we understand how much each feature contributes to the model's output.

Calculating SHAP values involves considering all possible combinations of features and determining the marginal contribution of each feature across these coalitions. For a given feature, the Shapley value represents its average marginal contribution across all possible alliances. This is achieved by calculating the difference in the model's output with and without the presence of a particular feature in a specific coalition.

B Characteristic factors

B.1 Network

New addresses (NA) are unique identifiers or cryptographic keys generated for each transaction or user in a specific cryptocurrency network that is the source or destination of funds. In other words, new addresses are newly created addresses that received the first deposit in a particular cryptocurrency. The change rate of NA is given by

$$\Delta NA_{t+1} = \frac{NA_{t+1} - NA_t}{NA_t}.$$

Active addresses (AA) refer to the number of unique addresses participating in transactions within a given period. They can also be understood as addresses that conduct one or more on-chain transactions on a given date. Active addresses are a key indicator of user engagement and network activity levels in a specific cryptocurrency network. The change rate of AA is given by

$$\Delta AA_{t+1} = \frac{AA_{t+1} - AA_t}{AA_t}.$$

Zero balance addresses (ZBA) are addresses in a cryptocurrency network that hold no funds or

⁷ For more information, please see <https://shap.readthedocs.io/en/latest/>.

have a zero balance, i.e., an address from which all assets for that crypto asset are transferred. Zero balance addresses occur for a variety of reasons. A typical scenario is that users empty their wallets by transferring all funds to other addresses, or users close their accounts or stop participating in the network. The change rate of ZBA is given by

$$\Delta ZBA_{t+1} = \frac{ZBA_{t+1} - ZBA_t}{ZBA_t}.$$

Active addresses ratio (AAR) is a reliable measure of cryptocurrency network activity and user engagement, providing a valuable reference for the network's level of participation and utilization. This ratio quantifies the proportion of actively used addresses relative to the total number of addresses in a specific cryptocurrency network. The ratio is computed by

$$AAR_{t+1} = \frac{AA_{t+1}}{\text{Total number of addresses with a balance}_{t+1}}.$$

Number of transactions (NT) refers to the total number of verified and recorded transactions that occur on a specific cryptocurrency network during a given period. This metric is a key metric for understanding and assessing the level of economic activity and user engagement within the cryptocurrency ecosystem. The change rate of NT is given by

$$\Delta NT_t = \frac{NT_t - NT_{t-1}}{NT_{t-1}}.$$

Transactions volume (TV) refers to the total value of transactions conducted within a given cryptocurrency network over a period, usually measured in units of cryptocurrencies exchanged or the corresponding U.S. dollar. Transactions volume represents the cumulative value of all transactions processed within a cryptocurrency network, which can provide a quantitative indicator of financial activity and market liquidity. The change rate of TV is given by

$$\Delta TV_{t+1} = \frac{TV_{t+1} - TV_t}{TV_t}.$$

The total fee (TFE) is the total amount of fees paid every day to use the blockchain. The change rate of TFE is given by

$$\Delta TFE_{t+1} = \frac{TFE_{t+1} - TFE_t}{TFE_t}.$$

The average transaction fee (ATF) is the average daily transaction cost. The change rate of ATF is given by

$$\Delta ATF_{t+1} = \frac{ATF_{t+1} - ATF_t}{ATF_t}.$$

Total addresses (TA) are all addresses that have ever created a point holding a specific cryptocurrency, including those holding it. The change rate of TA is given by

$$\Delta TA_{t+1} = \frac{TA_{t+1} - TA_t}{TA_t}.$$

The average time between transactions (ATBT) measure the average duration between one transaction block and the next, which is the average of the total time required to approve a block after the previous block in the blockchain. The change rate of ATBT is given by

$$\Delta ATBT_{t+1} = \frac{ATBT_{t+1} - ATBT_t}{ATBT_t}.$$

Created unspent transaction outputs (CUTO) refer to the total amount of transactions and funds generated in a blockchain transaction. The change rate of CUTO is given by

$$\Delta CUTO_{t+1} = \frac{CUTO_{t+1} - CUTO_t}{CUTO_t}.$$

Spent unspent transaction output (SUTO) is a previously created unspent transaction output and is used as an input to be transferred to one or more addresses. This metric is a key component of Bitcoin transactions and reflects the total number of unspent transaction outputs sent as transaction inputs to one or more addresses. The change rate of SUTO is given by

$$\Delta SUTO_{t+1} = \frac{SUTO_{t+1} - SUTO_t}{SUTO_t}.$$

Holding time of transacted coins (HTTC) measure how long the token was held before the transaction. The change rate of HTTC is given by

$$\Delta HTTC_{t+1} = \frac{HTTC_{t+1} - HTTC_t}{HTTC_t}.$$

B.2 Finance

The market value to realized value ratio (MRR) is used to evaluate the valuation of a cryptocurrency in relation to its historical on-chain transaction activity. It is the ratio of the market value of a cryptocurrency to its realized value. The ratio is given by

$$MRR_{t+1} = \frac{\text{Market value}_{t+1}}{\text{Average purchasing cost of each address}_{t+1}},$$

where the realized value is the average purchase cost of each address, which is calculated based on the price of all tokens last moved on the chain.

The network value to transaction ratio (NTR) is a key metric to assess cryptocurrencies' valuation and economic activity within their respective blockchain networks. It is the ratio of a cryptocurrency's market value to its transaction value and is given by

$$NTR_{t+1} = \frac{\text{Market value}_{t+1}}{\text{Transaction}_{t+1}}.$$

B.3 Ownership

Large holders netflow (LHN) measures the net change in cryptocurrency funds held by large investors or entities. This indicator indicates whether influential holders are accumulating or allocating their cryptocurrency assets, reflecting their market sentiment and potential market impact. LHN and its change rate are given by

$$LHN_{t+1} = \text{Large holders inflow}_{t+1} - \text{Large holders outflow}_{t+1},$$

$$\Delta LHN_{t+1} = \frac{LHN_{t+1} - LHN_t}{LHN_t}.$$

The change rates of large holders inflow and large holders outflow are given by

$$\Delta LHI_{t+1} = \frac{LHI_{t+1} - LHI_t}{LHI_t},$$

$$\Delta LHO_{t+1} = \frac{LHO_{t+1} - LHO_t}{LHO_t}.$$

B.4 Exchange

Net flows (NF) measure the net change in the amount of funds deposited or withdrawn from cryptocurrency exchanges. It is the difference between the inflow volume and outflow volume of

cryptocurrency exchange funds, i.e.,

$$NF_{t+1} = \text{Inflow volume}_{t+1} - \text{Outflow volume}_{t+1}.$$

The change rate of NF is given by

$$\Delta NF_{t+1} = \frac{NF_{t+1} - NF_t}{NF_t}.$$

Total flows (TF) measure the total amount of funds deposited and withdrawn from cryptocurrency exchanges. It is the sum of the inflow and outflow volume of funds from cryptocurrency exchanges, i.e.,

$$TF_{t+1} = \text{Inflow volume}_{t+1} + \text{Outflow volume}_{t+1}.$$

The change rate of TF is given by

$$\Delta TF_{t+1} = \frac{TF_{t+1} - TF_t}{TF_t}.$$

Netflow transaction count (NTC) is the difference between the inflow transaction count and outflow transaction count, that is

$$NTC_{t+1} = \text{Inflow transaction count}_{t+1} - \text{Outflow transaction count}_{t+1}.$$

NTC can comprehensively reflect the situation of users sending and withdrawing funds to the exchange and its change rate is given by

$$\Delta NTC_{t+1} = \frac{NTC_{t+1} - NTC_t}{NTC_t}.$$

B.6 Social

Google Trends (GT) of a cryptocurrency directly reflects investor attention to that cryptocurrency and does not use transaction data in the cryptocurrency market, which may be unreliable. Existing studies have found that investor attention may be related to the future returns of cryptocurrencies (Sockin and Xiong, 2019; Liu and Tsyvinski, 2021). The change rate of GT is given by

$$\Delta GT_{t+1} = \frac{GT_{t+1} - GT_t}{GT_t}.$$

B.7 Market, size, and momentum

Inspired by Fama and French (1993, 2012), we construct market, size, and momentum factors in the cryptocurrency market. The cryptocurrency market return is computed by weighting the market capitalization of the cryptocurrency and is given by

$$RM_t = \sum_{i=1}^{3940} \frac{MC_{i,t}}{MC_{MKT,t}} \times r_{i,t}$$

where RM_t represents the return of the cryptocurrency market on week t , $MC_{i,t}$ represents the market capitalization of the i th cryptocurrency on week t , $MC_{MKT,t}$ represents the total market capitalization on week t , and $MC_{MKT,t} = \sum_{i=1}^{3940} MC_{i,t}$, $r_{i,t}$ represents the return of the i th cryptocurrency on week t . Then, the market factor is measured by excess market return and is given by

$$RMRF_t = RM_t - RF_t,$$

where RF_t denotes the risk-free rate on week t .

The size and momentum factors are derived from market capitalization and the performance of cryptocurrency returns over the previous four weeks. To construct the size factor, we categorize the top 50% of cryptocurrencies by market capitalization as a large portfolio (Big), while the bottom 50%

form a small portfolio (Small). For the momentum factor, we use breakpoints at the 30th and 70th percentiles based on the previous four weeks of cryptocurrency returns. Three distinct investment portfolios emerge from this categorization: Up (top 30%), Medium (middle 40%), and Down (bottom 30%). Consequently, six value-weighted portfolios are formed at the intersection of two size and three momentum portfolios. We use SMB (small minus big) to measure the size factor, which is the difference between the average returns of three small cryptocurrency portfolios and three large cryptocurrency portfolios, that is

$$SMB = \frac{1}{3}(Small\ Up + Small\ Medium + Small\ Down) - \frac{1}{3}(Big\ Up + Big\ Medium + Big\ Down).$$

Similarly, we use UMD (up minus down) to measure the momentum factor, which is the difference between the average returns of two high prior return portfolios and two low prior return portfolios, that is

$$UMD = \frac{1}{2}(Big\ Up + Small\ Up) - \frac{1}{2}(Big\ Down + Small\ Down).$$

C Macroeconomic factors

C.1 Uncertainty indices

The economic policy uncertainty (EPU) index proposed by Baker et al. (2016) quantifies the level of uncertainty of economic policy in the U.S.⁸ The EPU return is given by

$$\Delta EPU_{t+1} = \frac{EPU_{t+1} - EPU_t}{EPU_t}.$$

The Twitter-based economic uncertainty (TEU) index derived by Baker et al. (2021) utilizes real-time Twitter data to measure economic uncertainty.⁹ Traditional economic indicators and surveys are lagging, and it is difficult to measure the current economic environment accurately. By leveraging Twitter data, the TEU index provides an updated and dynamic measure of economic uncertainty. The TEU return is given by

$$\Delta TEU_{t+1} = \frac{TEU_{t+1} - TEU_t}{TEU_t}.$$

C.2 Market indices

We select the S&P 500 (SPX) as a measure of the U.S. stock market. The SPX return is given by

$$\Delta SPX_{t+1} = \frac{SPX_{t+1} - SPX_t}{SPX_t}.$$

We select the volatility index (VIX) as a measure of the U.S. options market. The VIX return is given by

$$\Delta VIX_{t+1} = \frac{VIX_{t+1} - VIX_t}{VIX_t}.$$

We select the U.S. dollar index (USDXX) as a measure of global foreign exchange market. The USDXX return is given by

⁸ This index is downloaded from the website https://www.policyuncertainty.com/us_monthly.html.

⁹ This index is downloaded from the website https://www.policyuncertainty.com/twitter_uncert.html.

$$\Delta USDX_{t+1} = \frac{USDX_{t+1} - USDX_t}{USDX_t}.$$

C.3 Commodity prices

We select the gold price (GLD) in COMEX as a measure of global gold market. The GLD return is given by

$$\Delta GLD_{t+1} = \frac{GLD_{t+1} - GLD_t}{GLD_t}.$$

We select the crude oil price (WTI) in NYMEX as a measure of global crude oil market. The WTI return is given by

$$\Delta WTI_{t+1} = \frac{WTI_{t+1} - WTI_t}{WTI_t}.$$

C.4 Cryptocurrency index

Inspired by Aslanidis et al. (2022), we apply 5 keywords (Bitcoin, BTC, cryptocurrency, blockchain, and crypto) to Google Trends and thus construct the Google-Trends-based cryptocurrency index (GTCX).¹⁰

Table C.4.1

The top 10 search terms are most relevant to the keyword “Bitcoin.” The second column is the correlation coefficient between the search term and the keyword “Bitcoin,” and the third column is the search volume ratio of the search term and the keyword “Bitcoin.”

Rank	Correlation	Relative search volume	Search term
1	0.629	93.12%	bitcoin usd
2	0.623	82.12%	bitcoin price
3	0.465	94.08%	bitcoin rate
4	0.422	82.62%	current bitcoin
5	0.415	92.06%	bitcoin dollar
6	0.349	114.05%	mining bitcoin
7	0.250	94.05%	bitcoin currency
8	0.241	63.48%	current bitcoin value
9	0.235	93.75%	bitcoin trading
10	0.095	65.06%	how bitcoin works

Among the above keywords, “Bitcoin” is the most important keyword. This keyword is the most widely used search term by investors interested in the cryptocurrency market. On the other hand, Bitcoin’s market capitalization is the largest in the cryptocurrency market, and its influence on the market is also the largest. We have examined search terms that are highly relevant to this keyword. Table C.4.1 shows the 10 most relevant search terms and compares their search volume to “Bitcoin.” The top 5 are the most relevant but have lower search volumes. Although the search volume for “mining bitcoin” is high, its relevance is low. The bottom 4 have very low relevance and search

¹⁰ The Google Trends series is downloaded from the website <https://trends.google.com/trends/?geo=US>.

volume. The remaining 4 keywords have similar results. Thus, we do not add additional information to the above keywords.

Table C.4.2 shows the Google Trends correlation between 5 keywords. Due to the low correlation between the series, the GTCX is calculated using the weighted average of the Google Trends series as follows

$$GTCX_t = \sum_{i=1}^5 \frac{GT_{i,t}}{TGT_t} \times GT_{i,t}$$

where $GT_{i,t}$ represents the Google Trends time series of keyword i on day t , and TGT_t represents the sum of time series of all keywords on day t . The index returns on day $t + 1$ is given by

$$\Delta GTCX_{t+1} = \frac{GTCX_{t+1} - GTCX_t}{GTCX_t}.$$

Table C.4.2

The correlation of Google Trends series of 5 keywords.

	Bitcoin	BTC	cryptocurrency	crypto	blockchain
Bitcoin	1.000				
BTC	0.401	1.000			
cryptocurrency	0.333	0.044	1.000		
crypto	0.302	0.421	0.253	1.000	
blockchain	-0.084	0.182	0.164	0.272	1.000

D Subgroup analysis

In addition to utilizing out-of-sample R^2 for assessment, we employ the Diebold-Mariano (1995) test to conduct comparisons across various methods considering mega-cap, large-cap, and mid-cap cryptocurrencies. This test evaluates differences in out-of-sample predictive accuracy between two models. The results of the Diebold-Mariano test for mega-cap, large-cap, and mid-cap cryptocurrencies are provided in Table D.1, D.2, and D.3, respectively.

Consistent with the findings obtained from the out-of-sample R^2 analysis, our results from the Diebold-Mariano test confirm the effectiveness of using out-of-sample R^2 as a robust metric for model selection across different cryptocurrency market segment.

Table D.1

Comparison of weekly out-of-sample prediction for mega-cap cryptocurrencies using Diebold-Mariano test.

	OLS-3	PLS	LASSO	Ridge	ENet	RF	GBRT	KNN	NN1	NN2	NN3	NN4	NN5
OLS	-.0675	0.000	-.3.192*	0.001	-.3.192*	4.365*	4.081*	0.680	2.562*	2.388*	2.482*	3.194*	2.394*
OLS-3		0.675	-.2.516*	0.676	-.2.516*	5.040*	4.756*	1.356	3.237*	3.063*	3.157*	3.869*	3.069*
PLS			-.3.192*	0.001	-.3.192*	4.365*	4.081*	0.680	2.562*	2.388*	2.482*	3.194*	2.394*
LASSO				3.192*	0.000	7.556*	7.273*	3.872*	5.754*	5.579*	5.674*	6.386*	5.585*
Ridge					-.3.192*	4.364*	4.081*	0.680	2.562*	2.387*	2.482*	3.194*	2.393*
ENet						7.556*	7.273*	3.872*	5.754*	5.579*	5.674*	6.386*	5.585*
RF							-.0.284	-.3.684*	-.1.802*	-.1.977*	-.1.882*	-.1.171	7.502*
GBRT								-.3.401*	-.1.519	-.1.693	-.1.599	-.0.887	-.1.688
KNN									1.882*	1.708*	1.802*	2.514*	1.713*
NN1										-.0.174	-.0.080	0.632	-.0.169
NN2											0.094	0.806	0.006
NN3												0.712	-.0.089
NN4													-.0.081

Table D.2

Comparison of weekly out-of-sample prediction for large-cap cryptocurrencies using Diebold-Mariano test.

	OLS-3	PLS	LASSO	Ridge	ENet	RF	GBRT	KNN	NN1	NN2	NN3	NN4	NN5
OLS	−1.309	0.000	−1.419	0.002	−1.419	3.383*	2.349*	1.645*	1.843*	1.801*	0.985	0.950	1.961*
OLS-3		1.309	−0.110	1.311	−0.110	4.692*	3.658*	2.955*	3.152*	3.110*	2.295*	2.259*	3.270*
PLS			−1.419	0.002	−1.419	3.383*	2.349*	1.645*	1.843*	1.801*	0.985	0.950	1.961*
LASSO				1.421	0.000	4.802*	3.768*	3.064*	3.262*	3.220*	2.405*	2.369*	3.380*
Ridge					−1.421	3.381*	2.347*	1.643*	1.841*	1.799*	0.984	0.948	1.959*
ENet						4.802*	3.768*	3.064*	3.262*	3.220*	2.405*	2.369*	3.380*
RF							−1.034	−1.738*	−1.540	−1.582	−2.398*	−2.433*	4.802*
GBRT								−0.703	−0.506	−0.548	−1.363	−1.399	−0.388
KNN									0.198	0.155	−0.660	−0.695	0.315
NN1										−0.042	−0.857	−0.893	0.118
NN2											−0.815	−0.851	0.160
NN3												−0.036	0.975
NN4													1.011

Table D.3

Comparison of weekly out-of-sample prediction for mid-cap cryptocurrencies using Diebold-Mariano test.

	OLS-3	PLS	LASSO	Ridge	ENet	RF	GBRT	KNN	NN1	NN2	NN3	NN4	NN5
OLS	−0.723	0.000	−0.701	0.000	−0.701	2.509*	1.692	1.710*	1.613	1.991*	1.678	1.417	1.908*
OLS-3		0.723	0.022	0.723	0.022	3.232*	2.416*	2.434*	2.336*	2.714*	2.402*	2.140*	2.632*
PLS			−0.701	0.000	−0.701	2.509*	1.692	1.710*	1.613	1.991*	1.678	1.417	1.908*
LASSO				0.701	0.000	3.209*	2.393*	2.411*	2.314*	2.692*	2.379*	2.118*	2.609*
Ridge					−0.701	2.509*	1.692	1.710*	1.613	1.991*	1.678	1.417	1.908*
ENet						3.209*	2.393*	2.411*	2.314*	2.692*	2.379*	2.118*	2.609*
RF							−0.186	−0.798	−0.896	−0.518	−0.830	−1.092	3.206*
GBRT								0.018	−0.080	0.299	−0.014	−0.276	0.216
KNN									−0.098	0.281	−0.032	−0.294	0.198
NN1										0.378	−0.066	−0.196	0.296
NN2											−0.313	−0.574	−0.083
NN3												−0.262	0.230
NN4													0.492

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E Equal-weighted portfolios

Table E.1 reports the out-of-sample performance for equal-weighted long-short and long-only portfolios, while Figure E.1 shows the dynamics of cumulative returns for both long-short and long-only equal-weighted portfolios. The dynamics of cumulative returns for neural network equal-weighted portfolios are provided in Figure E.2.

Table E.1

The out-of-sample performance for equal-weighted portfolios on the full sample.

	Mean	Std	Sharpe	Skewness	Kurtosis	MaxDD
Long-short						
OLS	16.462	13.232	430.971	2.121	8.048	99.060
OLS-3	7.354	10.020	254.242	2.551	14.094	114.486
PLS	19.169	15.382	431.715	2.152	6.516	99.565
LASSO	14.078	11.671	417.854	0.929	0.367	107.477
ENet	14.163	11.600	422.951	0.909	0.361	107.426
Ridge	21.966	14.247	534.091	0.963	0.963	94.030
RF	33.774	20.716	564.265	1.267	1.728	96.069
GBRT	35.142	21.364	569.809	1.055	0.747	94.894
KNN	28.897	17.284	579.154	1.049	0.432	93.382
NN1	33.983	21.177	556.172	1.214	1.121	95.287
NN2	29.696	18.912	543.956	1.467	2.459	94.746
NN3	27.813	19.072	505.193	1.349	1.854	95.866
NN4	30.368	20.043	524.875	1.375	2.845	94.836
NN5	26.894	18.603	500.799	1.300	1.937	98.137
Long-only						
OLS	12.587	17.602	247.714	0.980	1.054	183.072
OLS-3	5.014	11.618	149.500	−0.274	2.223	213.545
PLS	11.682	16.305	248.192	1.093	2.107	182.927
LASSO	8.965	18.979	163.624	−0.320	6.659	218.883
ENet	9.256	19.060	168.230	−0.365	6.237	224.020
Ridge	14.768	21.679	235.976	−0.135	5.161	219.573
RF	26.146	26.219	345.446	1.618	2.885	110.092
GBRT	25.986	28.013	321.339	2.044	5.253	108.670
KNN	21.140	26.951	271.724	0.578	4.132	180.234
NN1	26.299	28.149	323.640	1.556	2.290	113.333
NN2	25.024	24.779	349.828	1.738	3.539	106.489
NN3	22.693	27.630	284.519	0.279	3.100	181.097
NN4	25.335	32.698	268.403	−0.212	7.611	201.845
NN5	25.099	26.840	323.945	1.558	2.786	110.735

Notes: This table reports the out-of-sample performance measures for both long-short and long-only portfolios constructed by all machine learning models. The portfolios are equal-weighted and based on the full sample. All measures are derived from the out-of-sample returns from January 2014 to March 2023. “Mean” represents the average predicted weekly return (%). “Std” denotes the standard deviation of the predicted weekly returns (%). “Sharpe” represents the annualized Sharpe ratio (%).

“MaxDD” denotes the maximum drawdown (%) of the portfolio.

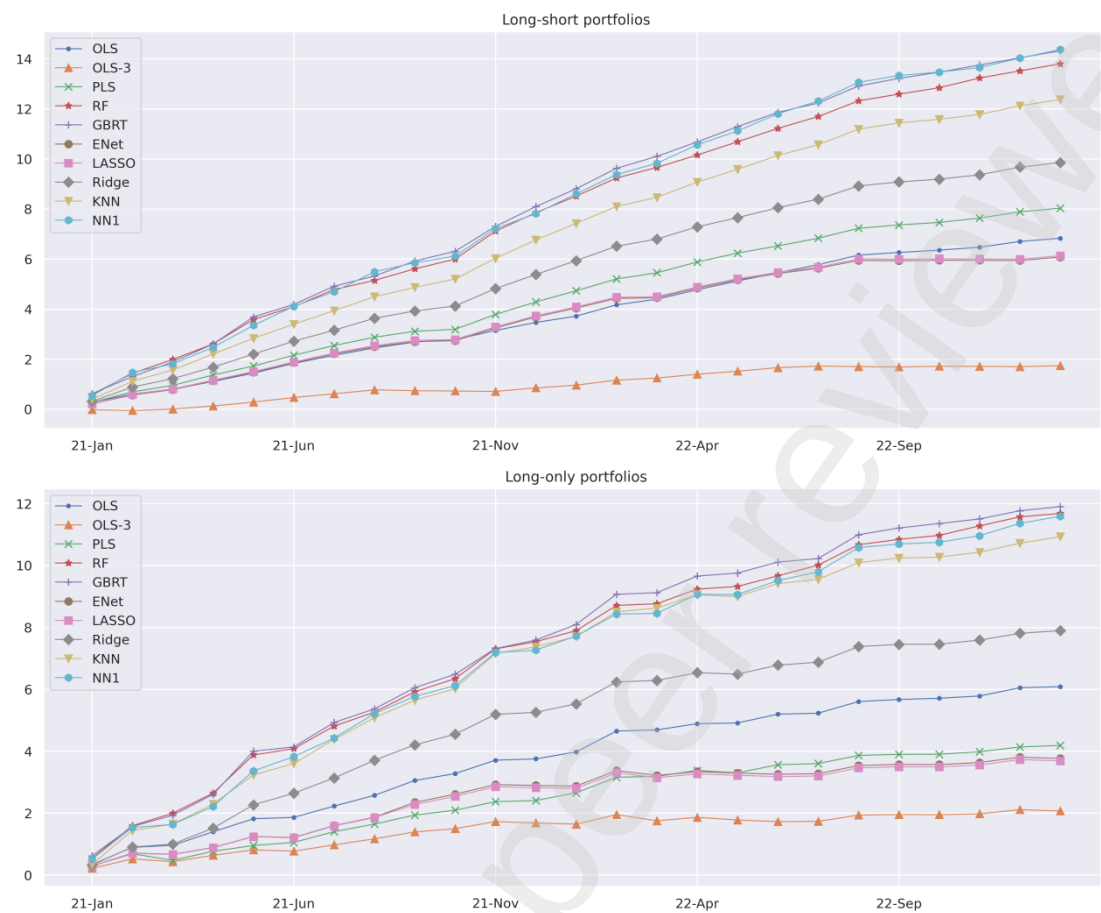


Figure E.1. Cumulative returns of equal-weighted portfolios based on the full sample.

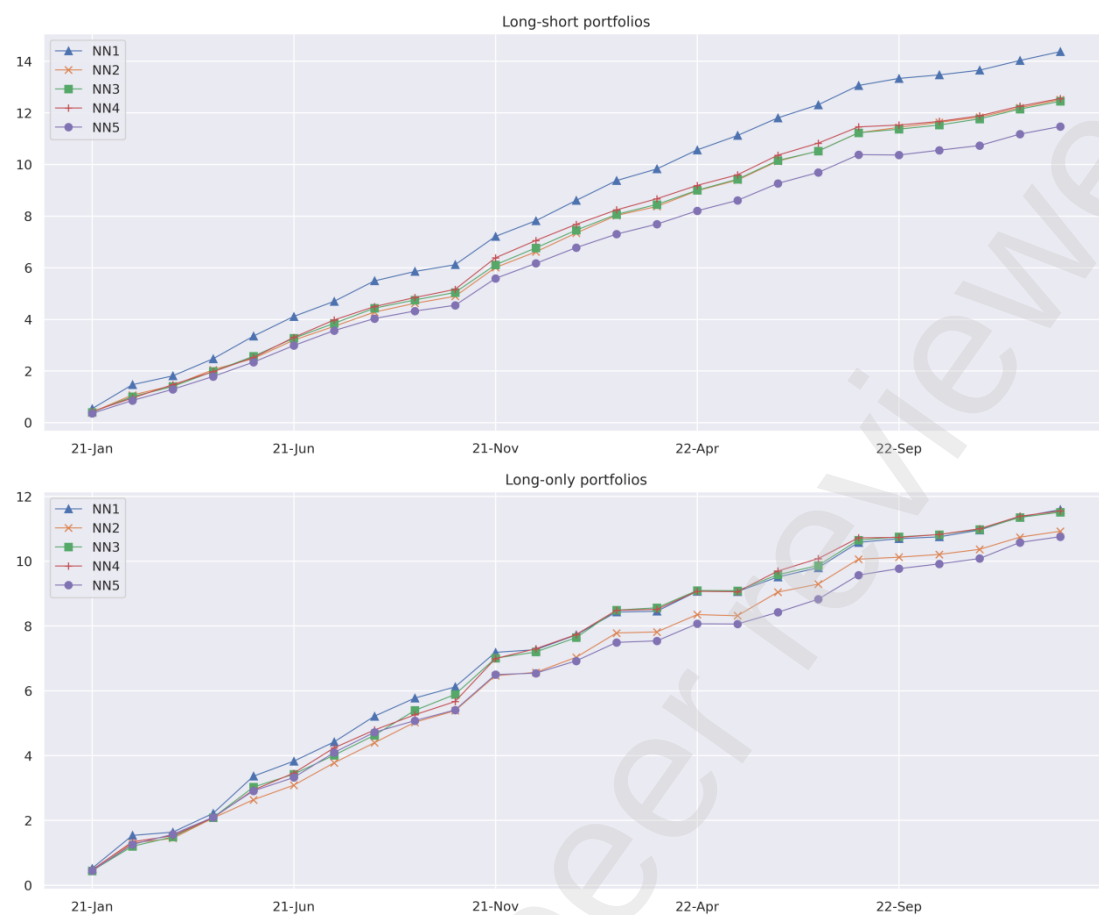


Figure E.2. Cumulative returns of neural network portfolios (equal-weighted) based on the full sample.